

Unit 1

LC 1: Introduction to Data Communication

1.1 DATA COMMUNICATIONS

The term telecommunication means communication at a distance.

What is a data?

The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data.

What is a Data communications ?

Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Characteristics for effective data communication:

delivery, accuracy, timeliness and jitter.

1. **Delivery:** The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.
2. **Accuracy:** The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.
3. **Timeliness:** The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called Real-Time Transmission.
4. **Jitter:** Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets. For example, let us assume that video packets are sent every 30ms. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result.

Data Communication system components:

A data communications system has five components (see figure 1.1)

1. **Message:** The **message** is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio and video.
2. **Sender:** The **sender** is the device that sends the data message. It can be a computer, workstation, telephone handset, and so on.
3. **Receiver:** The **receiver** is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
4. **Transmission medium:** The **transmission medium** is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
5. **Protocol:** A **protocol** is a set of rules that govern data communications. It represents an

agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking English cannot be understood by a person who speaks only kannada.

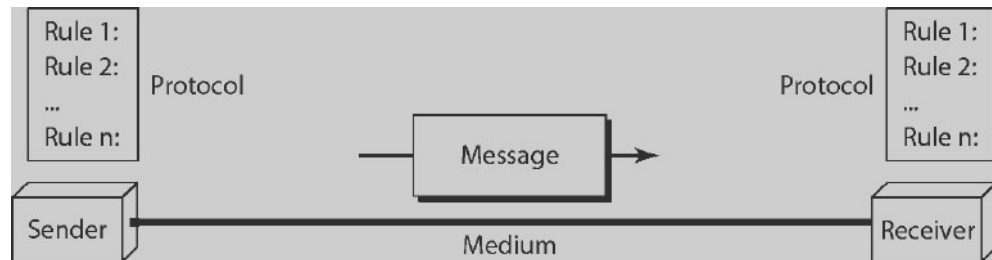


Fig: 1.1

Types of Data Representation:

Text

In data communication, text is represented as a bit pattern, a sequence of bits (0s or 1s). Different sets of bit patterns have been designed to represent text symbols. Each set is called a **code**, and the process of representing symbols is called coding. Today, the prevalent coding system is called **Unicode**, which uses 32 bits to represent a symbol or character used in any language in the world. The **American Standard Code for Information Interchange (ASCII)** developed some decades ago in the United States, now constitutes the first 127 characters in Unicode and is also referred to as **Basic Latin**.

Numbers

Numbers are also represented by bit patterns. However, a code such as ASCII is not used to represent numbers; the number is directly converted to a binary number to simplify mathematical operations.

Images

Images are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is a small dot. The size of the pixel depends on the resolution. For example, an image can be divided into 1000 pixels or 10,000 pixels. In the second case, there is a better representation of the image (better resolution), but more memory is needed to store the image.

After an image is divided into pixels, each pixel is assigned a bit pattern. The size and the value of the pattern depend on the image. For an image made of only black-and-white dots (e.g., a chessboard), a 1-bit pattern is enough to represent a pixel.

If an image is not made of pure white and pure black pixels, you can increase the size of the bit pattern to include gray scale. For example, to show four levels of gray scale, you can use 2-bit patterns. A black pixel can be represented by 00, a dark gray pixel by 01, a light gray pixel by 10, and a white pixel by 11.

There are several methods to represent colour images. One method is called **RGB**, so called because each colour is made of a combination of three primary colours: red, green, and blue. The intensity of each colour is measured, and a bit pattern is assigned to it. Another method is called **YCM**, in which a colour is made of a combination of three other primary colours: yellow, cyan, and magenta.

Audio

Audio refers to the recording or broadcasting of sound or music. Audio is by nature different from

text, numbers, or images. **It is continuous**, not discrete. Even when we use a microphone to change voice or music to an electric signal, we create a continuous signal.

Video

Video refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity (e.g., by a TV camera), or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion. Again we can change video to a digital or an analog signal.

Data Flow

Communication between two devices can be simplex, half-duplex, or full-duplex as shown in figure 1.2

Simplex

In **simplex mode**, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive (see figure 1.2a).

Keyboards and traditional monitors are examples of simplex devices. The keyboard can only introduce input; the monitor can only accept output. The simplex mode can use the entire capacity of the channel to send data in one direction.

Half-Duplex

In **half-duplex mode**, each station can transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa (see figure 1.2b).

The half-duplex mode is like a one-lane road with traffic allowed in both directions. When cars are travelling in one direction, cars going the other way must wait.

In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.

Examples: Walkie-talkie and CB (citizens band) radios are both half-duplex systems.

The half-duplex mode is used in cases where there is no need for communication in both directions at the same time; the entire capacity of the channel can be utilized for each direction.

Full-Duplex

In **full duplex mode** (also called **duplex**), both stations can transmit and receive simultaneously (see figure 1.2c).

The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time.

In full-duplex mode, signals going in one direction share the capacity of the link with signals going in the other direction.

This sharing can occur in two ways; Either the link must contain two physically separate transmission paths, one for sending and the other for receiving; or the capacity of the channel is divided between signals travelling in both directions.

One common example of full-duplex communication is the telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time.

The full-duplex mode is used when communication in both directions is required all the time. The capacity of the channel, however, must be divided between the two directions.

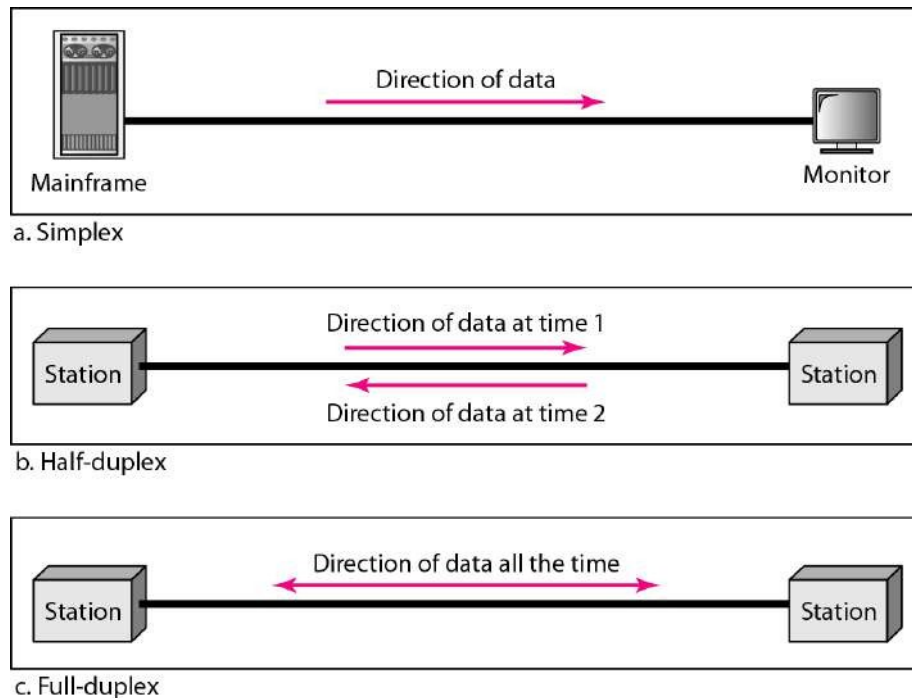


Fig:1.2

1-2 NETWORKS

A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Distributed Processing

A Network is a set of devices/nodes connected by communication links, each device capable of sending or receiving data generated by other nodes.

Most networks use Distributed Processing, in which the task is divided among multiple computers

Network Criteria

The most important criteria are Performance, Reliability and Security

Performance: Is often evaluated by two networking metrics, **Throughput and delay.**

Performance depends on number of factors like transit time (time taken for a msg to travel from one device to another), no. of users, type of transmission medium, efficiency of hardware and software.

Reliability: It is the accuracy of delivery, also measured by frequency of failure

Security: Protecting data from unauthorized access and implementing policies and procedures for recovery.

Physical Structures- Types of connections

A network is a set of communication devices connected by media links. .

The Possible types of connections are **Point-to-Point and Multipoint**

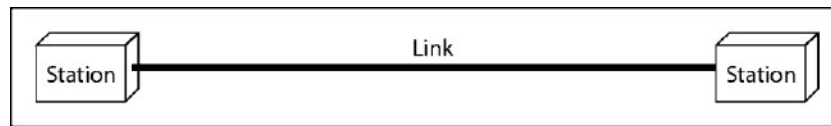
- * In a point-to-point connection, two and only two devices are connected by a dedicated link .
- * The entire capacity of the link is reserved for transmission between the two devices.
- * The links can be of microwave or satellite or a cable connecting the two ends.

Ex: Changing television channels by remote control.

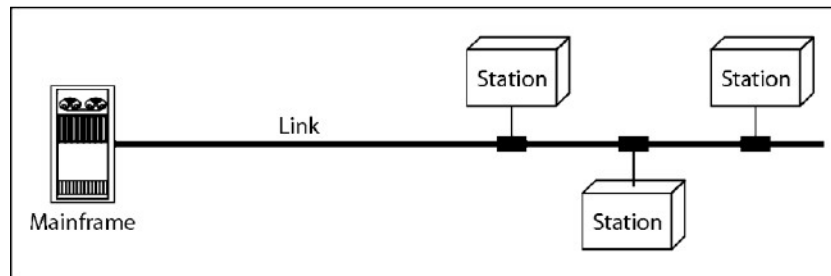
In a multipoint connection, three or more devices share a link.

Ex: LAN

Types of connections: point-to-point and multipoint



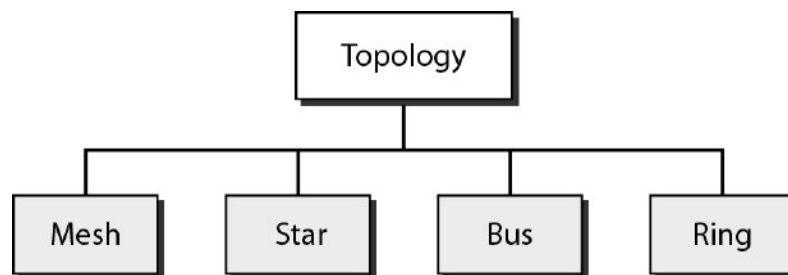
a. Point-to-point



b. Multipoint

LC 2: Network Topologies Layering and Protocols

Physical Structures- Physical Topology

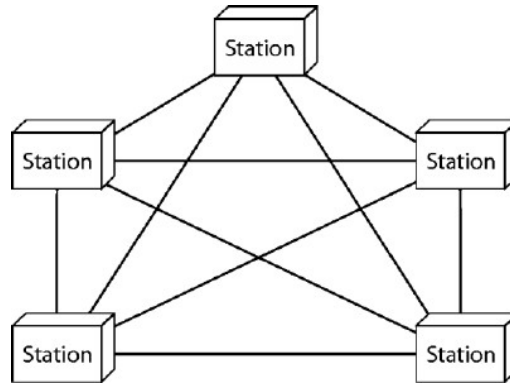


Topology refers to the physical or logical arrangement of a network.

i.e two or more devices connect to a link, two or more links form a topology.

Four basic topologies are: mesh, star, bus and ring topology.

- * In a **Mesh topology**, every device has a dedicated point-to-point link to every other device
- * A mesh network with n nodes need $n(n-1)$ physical links and $n(n-1)/2$ for duplex links.
- * To accommodate the links, every device on the mesh must have $(n-1)$ I/O ports.



Advantages of Mesh Topology:

- * Use of dedicated links eliminates traffic problems that can occur with shared links.
- * It is Robust, If one link becomes unusable, the entire system will not fail.
- * Enables Privacy or security, message travels along a dedicated line and only intended recipient sees it.
- * Point-to-point links make fault identification easy. Traffic can be routed to avoid links with suspected problems.

Disadvantages of Mesh topology:

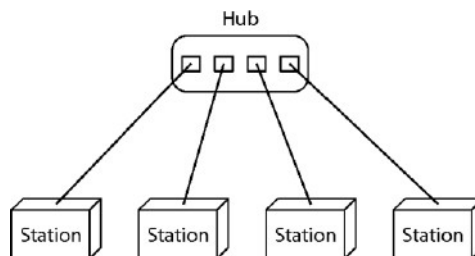
- * Installation and reconnection are difficult, b'coz every device must be connected to every other device.
- * Sheer bulk of wiring require more accommodation space.
- * The hardware required to connect each link(I/O Ports and cable) can be expensive.

Ex: Connection of telephone regional offices.

Star topology

- * Each device has a dedicated point-to-point link only to a central controller, usually called a hub.
- * The controller acts as an exchange: If one device wants to send data to another, it sends data to the controller, which then relays to the other device.
- * Therefore no direct traffic between devices.

A star topology connecting four stations



Advantages of STAR Topology

- * Less Expensive, each device needs only one port and one link.
- * Easy to install and reconfigure.
- * Less cabling & additions, moves and deletions involve only one connection
- * Is robust: If one link fails only that is affected, others remain active.
- * As long as hub is working, it can be used to monitor link problems.

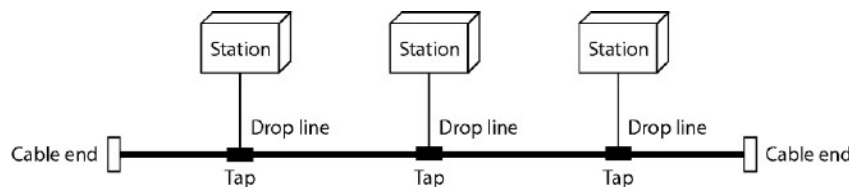
Disadvantages of STAR Topology:

- * Dependency of the whole topology on one single point. If the hub goes down, the whole system is dead.
- * Although a STAR requires less cabling than a mesh, it requires more cabling than some other topologies.

Ex: High-speed LAN's.

Bus Topology:

- * is Multipoint. One Cable links all devices in a network.
- * Nodes are connected to the bus cable by drop lines and taps.
- * There is a limit on the number of taps a bus can support and the distance between the taps.
- * As a signal travels along the backbone, some of its energy is transformed into heat, thereby weakening the signal as it travels farther and farther.



Advantages of Bus Topology

- *Ease of installation.
- *Uses less cabling than Mesh or Star topologies.

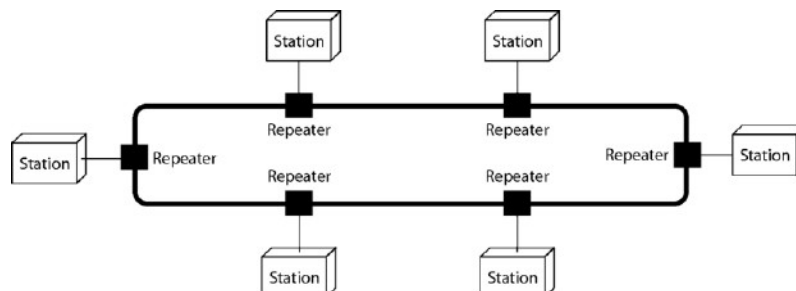
Disadvantages of Bus Topology

- * Difficult reconnection and fault isolation
- * Difficult to add new devices as it may require modification or replacement of the backbone.
- * A fault in the bus cable stops all transmission. The damaged area reflects signals back in the originating direction, creating noise in both directions.

Ex: Ethernet LAN's

In a Ring Topology

- * Each device has a dedicated point-to-point connection with only the two devices on either side of it.
- * A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- * Each device in the ring incorporates a repeater
- * when a device receives a signal intended for another device, the repeater regenerates the bits and passes them along.



Advantages of Ring Topology

- * Relatively easy to install and reconfigure.

- * Fault isolation is simplified, If one device does not receive a signal within a specified period, it can issue an alarm, the alarm alerts the network operator to the problem location.

Disadvantages

- * Unidirectional traffic can be a disadvantage, a break in the ring can disable the entire network.
- * This can be solved by using a dual ring or a switch closing the break.

PROTOCOLS AND STANDARDS

In this section, we define two widely used terms: protocols and standards. First, we define protocol, which is synonymous with rule. Then we discuss standards, which are agreed-upon rules.

A **protocol** is a set of rules that governs data communication;

The key elements of a protocol are syntax, semantics, and timing.

Syntax: Refers to the structure or format of data. Ex: A simple protocol might expect first 8 bits to be address of sender, next 8 bits to be address of receiver and the rest the message.

Semantics: Refers to the meaning of each section of bits. Ex: Address identify the route to be taken or the final destination

Timing: Refers to two things, When data should be sent and how fast they can be sent. Ex: Sender produces data at 100Mbps, Receiver processes data at 1Mbps, the transmission channel gets overloaded and data will be lost.

Standards are necessary to ensure that products from different manufacturers can work together as expected.

Data communications standards fall into two categories

de facto (Meaning —by fact or —by convention). Those standards have not approved by an organized body but have been adopted as standards through wide spread use.

de jure (Meaning —by law or —by regulation). Those standards have been legislated by an recognized body.

LC 3: Reference Model: OSI model, TCP/IP Reference model, Addressing.

Network Models

The two best known models are the OSI model and the Internet Model.

OSI Model defines a seven layer network.

Internet model defines a five layer network.

THE OSI MODEL

Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards.

An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.

An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.

The purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software.

The OSI model is not a protocol; it is a model for understanding and designing a network architecture that is flexible, robust, and interoperable.

ISO is the organization.

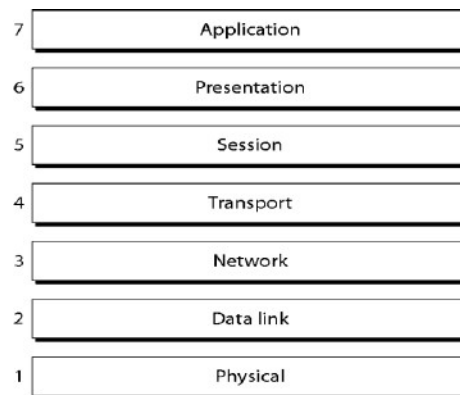
OSI is the model.

OSI is a layered model for the design of network systems that allows communication between different systems regardless of their architecture.

OSI model is composed of seven ordered layers: Physical, Data link, Network, Transport, Session, Presentation, Application.

Each layer uses the services provided by the layer below it and provides certain services for the layer above it.

Seven layers of the OSI model

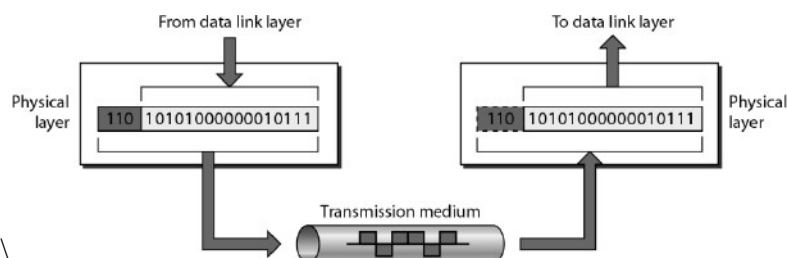


LAYERS IN THE OSI MODEL

Physical Layer

The physical layer coordinates the functions required to transmit a bit stream over a physical medium. It deals with the mechanical and electrical specifications of the interface and transmission medium. It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to occur.

The physical layer is responsible for movement of individual bits from one hop (node) to the next.



The physical layer is also concerned with the following functions:

Physical characteristics of interfaces and medium.

Representation of bits.

Data Rate

Synchronization of bits

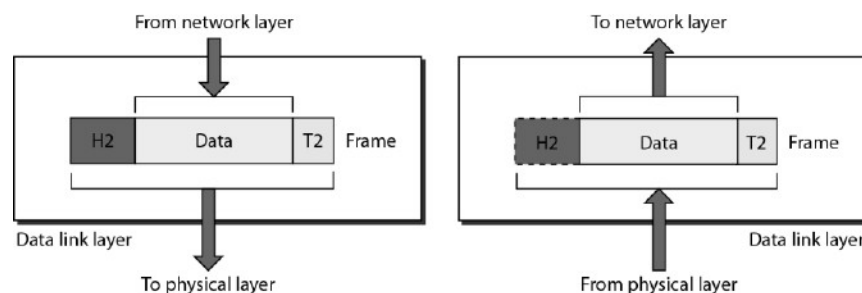
Line configuration
Physical topology
Transmission mode

1. **Physical characteristics of interfaces and medium:** The physical layer defines the characteristics of the interface between the devices and the transmission medium. It also defines the type of transmission medium.
2. **Representation of bits:** The physical layer data consists of a stream of **bits** (sequence of 0s or 1s) with no interpretation. To be transmitted, bits must be encoded into signals – electrical or optical. The physical layer defines the type of **encoding** (how 0s and 1s are changed to signals).
3. **Data rate:** The transmission rate – the number of bits sent each second – is also defined by the physical layer. In other words, the physical layer defines the duration of a bit, which is how long it lasts.
4. **Synchronization of bits:** The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level. In other words, the sender and the receiver clocks must be synchronized.
5. **Line configuration:** The physical layer is concerned with the connection of devices to the media. In a point-to-point configuration, two devices are connected through a dedicated link. In a multipoint configuration, a link is shared among several devices.
6. **Physical topology:** The physical topology defined how devices are connected to make a network. Devices can be connected by using a mesh topology (every device is connected to every other device), a star topology (devices are connected through a central device), a ring topology (each device is connected to the next, forming a ring), a bus topology (every device is on a common link), or a hybrid topology (this is a combination of two or more topologies).
7. **Transmission mode:** The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex. In simplex mode, only one device can send; the other can only receive. The simplex mode is a one-way communication. In the half-duplex mode, two devices can send and receive, but not at the same time. In a full-duplex (or simply duplex) mode, two devices can send and receive at the same time.

Data Link Layer

The **data link layer** transforms the physical layer, a raw transmission facility, to a reliable link. It makes the physical layer appear error-free to upper layer (network layer).

The data link layer is responsible for moving frames from one hop (node) to the next.



The data link layer is responsible for delivering data units from one station to the next without errors.

It's also responsible for the following:

Framing
Physical Addressing
Flow control
Error control
Access control

1. **Framing:** The data link layer divides the stream of bits received from the network layer into manageable data units called **frames**.
2. **Physical addressing:** If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame. If the frame is intended for a system outside the sender's network, the receiver address is the address of the device that connects to the network to the next one.
3. **Flow control:** If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.
4. **Error control:** The data link layer adds reliability to the physical layer by adding mechanism to detect and retransmit damaged or lost frames. It also uses a mechanism to recognize duplicate frames. Error control is normally achieved through a trailer added to the end of the frame.
5. **Access control:** When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.

The data link layer is responsible for moving frames from one hop (node) to the next.

As the below figure shows, communication at the data link layer occurs between two adjacent nodes. To send data from A to F, three partial deliveries are made.

First, the data link layer at A sends a frame to the data link layer at B (a router).

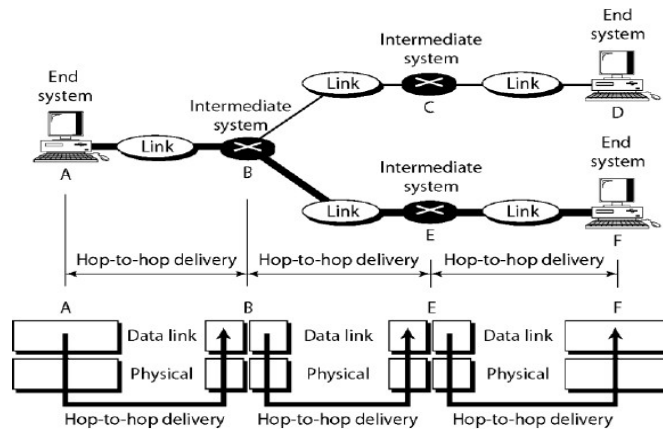
Second, the data link layer at B sends a new frame to the data link layer at E.

Finally, the data link layer at E sends a new frame to the data link layer at F. Note that the frames that are exchanged between the three nodes have different values in the headers.

The frame from A to B has B as the destination address and A as the source address.

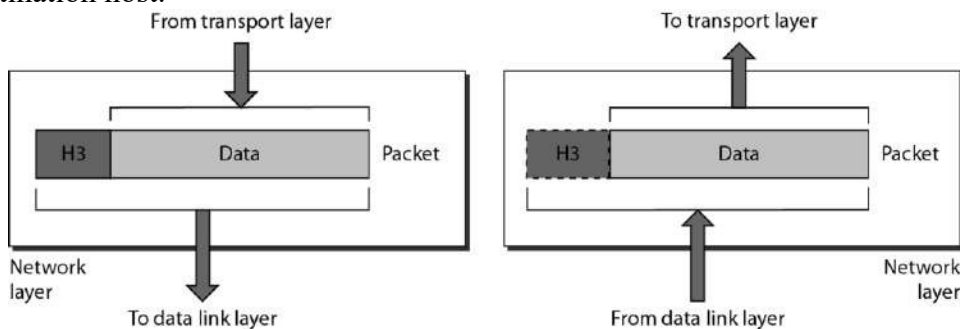
The frame from B to E has E as the destination address and B as the source address.

The frame from E to F has F as the destination address and E as the source address. The values of the trailers can also be different if error checking includes the header of the frame



Network Layer

The **network layer** is responsible for the delivery of individual packets from the source host to the destination host.

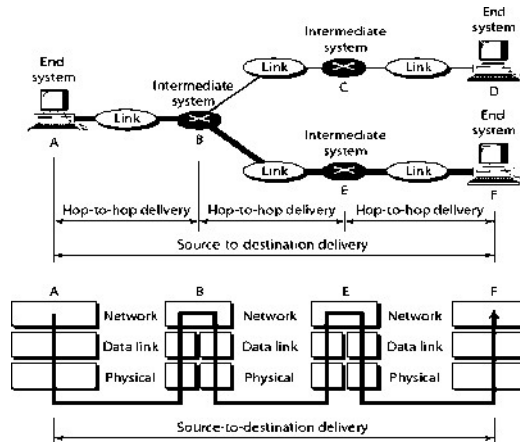


. Other responsibilities include:

Logical addressing
Routing

1. **Logical addressing:** The physical addressing implemented by the data link layer handles the addressing problem locally. If a packet passes the network boundary, we need another addressing system to help distinguish the source and destination systems. The network layer adds a header to the packet coming from the upper layer that, among other things, includes the logical addresses of the sender and receiver.
2. **Routing:** When independent networks or links are connected to create internetworks (network of networks) or a large network, the connecting devices (called routers or switches) route or switch the packets to their final destination. One of the functions of the network layer is to provide this mechanism.

The network layer is responsible for the delivery of individual packets from the source host to the destination host.



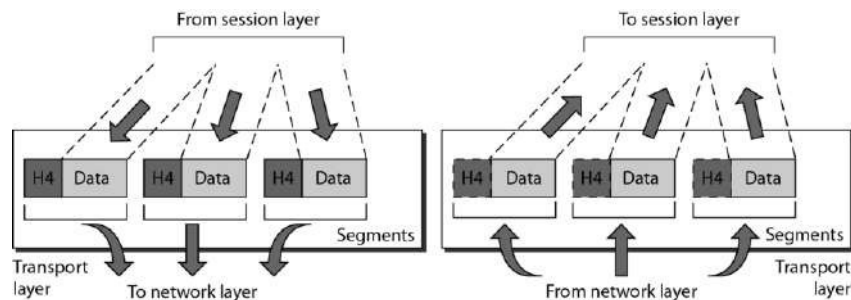
As the figure shows, now we need a source-to-destination delivery. The network layer at A sends the packet to the network layer at B. When the packet arrives at router B, the router makes a decision based on the final destination (F) of the packet. As we will see in later chapters, router B uses its routing table to find that the next hop is router E. The network layer at B, therefore, sends the packet to the network layer at F.

Transport Layer

The transport layer is responsible for the process-to-process delivery of the entire message

A process is an application program running on a host, whereas the network layer oversees **source-to-destination delivery** of individual packets.

The transport layer, on the other hand, ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level.



. Other responsibilities include:

- Service-point addressing**
- Segmentation and reassembly**
- Connection control**
- Flow control**
- Error Control**

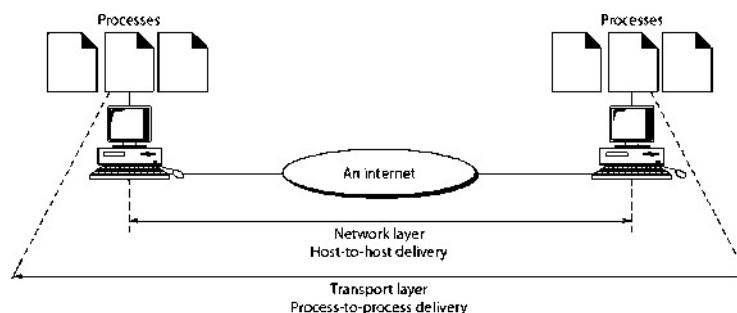
1. **Service-point addressing:** Computer often runs several programs at the same time. For this reason, source-to-destination delivery means delivery not only from one computer to the next but also from a specific process (running program) on one computer to a specific process (running program) on the other. The transport layer header must therefore include a type of address called a service-point address (or port address). The network layer gets each packet to the correct process on that computer.
2. **Segmentation and reassembly:** A message is divided into transmittable segments, with each segment containing a sequence number. These numbers enable the transport layer to reassemble the

message correctly upon arriving at the destination and to identify and replace packets that were lost in transmission.

3. **Connection control:** The transport layer can be either connectionless or connection-oriented. A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine. A connection-oriented transport layer makes a connection with the transport layer at the destination machine first before delivering the packets. After all the data are transferred the connection is terminated.
4. **Flow control:** Like the data link layer, the transport layer is responsible for flow control. However, flow control at this layer is performed end to end rather than across a single link.
5. **Error control:** Like the data link layer, the transport layer is responsible for **error control**. However, error control at this layer is performed process-to-process rather than across a single link. The sending transport layer makes sure that the entire message arrives at the receiving transport

layer without error (damage, loss, or duplication). Error correction is usually achieved through retransmission.

**The transport layer is responsible for the delivery
of a message from one process to another.**



Session Layer

It establishes, maintains, and synchronizes the interaction among communicating systems.

The Session layer is the network *dialog controller*. Other responsibilities include

Dialog control

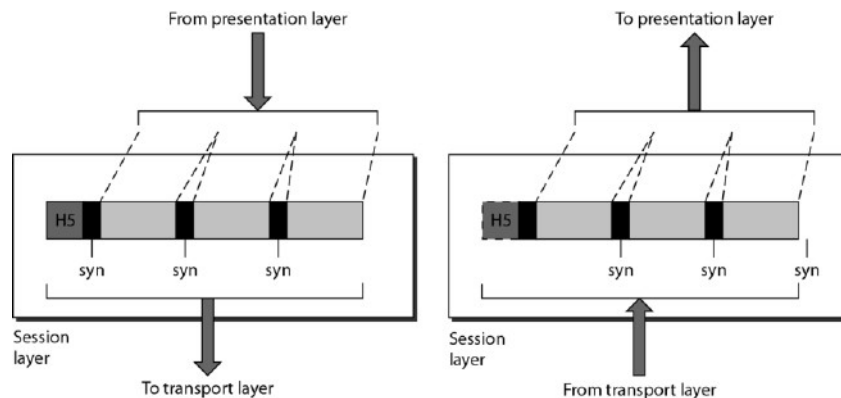
Synchronization: Allowing a process to add check points or synchronization points, to a stream of data

Specific responsibilities of the session layer include the following:

1. **Dialog control:** The session layer allows two systems to enter into a dialog. It allows the communication between two processes to take place in either half-duplex (one way at a time) or full-duplex (two ways at a time) mode.
2. **Synchronization:** The session layer allows a process to add checkpoints, or **synchronization points**, to a stream of data. For example, if a system is sending a file of 2000 pages, it is advisable to insert checkpoints after every 100 pages to ensure that each 100-page unit is received and acknowledged independently. In this case, if a crash happens during the transmission of page 523;

the only pages that need to be resent after system recovery are pages 501 to 523. Pages previous to 501 need not be resent. Figure 2.12 illustrates the relationship of the session layer to the transport and presentation layers.

The session layer is responsible for dialog control and synchronization.



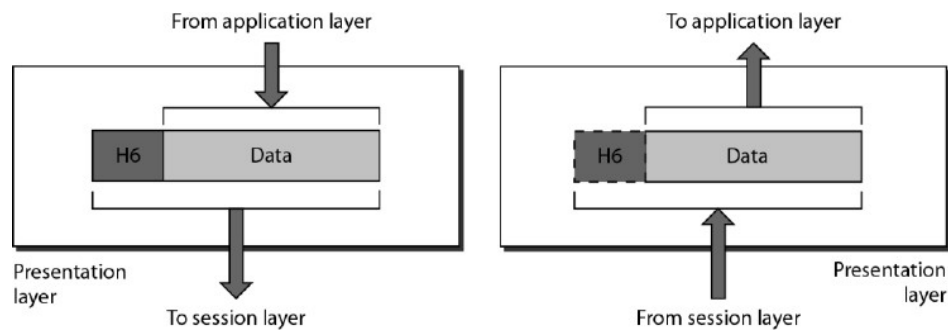
Presentation Layer

Presentation layer is concerned with the syntax and semantics of the information exchanged between two systems. Other responsibilities include:

Translation
Encryption
Compression

1. **Translation:** The processes (running program) in two systems are usually exchanging information in the form of character strings, numbers, and so on. The information must be changed to bit streams before being transmitted. Because different computers use different encoding systems, the presentation layer is responsible for interoperability between these different encoding methods. The presentation layer at the sender changes the information from its sender-dependent format into a common format into its receiver-dependent format.
2. **Encryption:** To carry sensitive information, a system must be able to ensure privacy. Encryption means that the sender transforms the original information to another form and sends the resulting message out over the network. Decryption reverses the original process to transform the message back to its original form.
3. **Compression:** Data compression reduces the number of bits contained in the information. Data compression becomes particularly important in the transmission of multimedia such as text, audio, and video.

The presentation layer is responsible for translation, compression, and encryption.

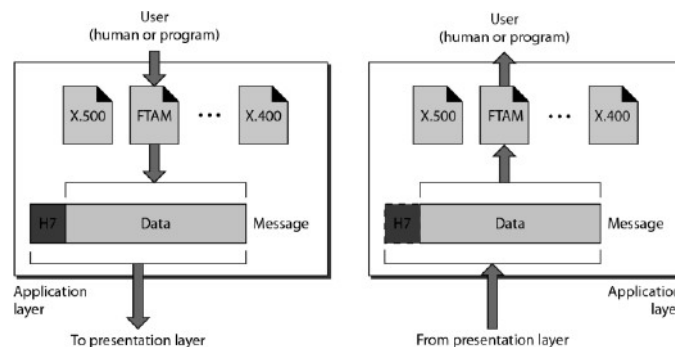


Application Layer

Of the many application services available, the figure shows only three: X.400 (message-handling services), X.500 (directory services), and file transfer, access, and management (FTAM). The user in this example employs X.400 to send an e-mail message.

The application layer enables the user to access the network.

It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management and other distributed information services.



The application layer is responsible for providing services to the user.

Specific services provided by application layer include:

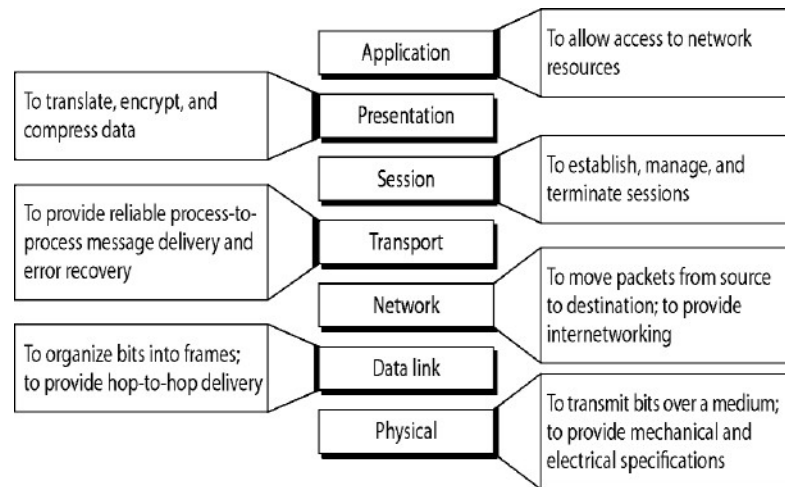
- Network virtual terminal
- File transfer, access and management
- Mail services
- Directory services

1. **Network virtual terminal:** A network virtual terminal is a software version of a physical terminal, and it allows a user to log on to a remote host. To do so, the application creates a software emulation of a terminal at the remote host. The user's computer talks to the software terminal which, in turn, talks to the host, and vice versa. The remote host believes it is communicating with one of its own terminals and allows the user to log on.
2. **File transfer, access, and management:** This application allows a user to access files in a remote host (to make changes or read data), to retrieve files from a remote computer for use in the local

computer, and to manage or control files in a remote computer locally.

3. **Mail services:** This application provides the basis for e-mail forwarding and storage.
4. **Directory services:** This application provides distributed database sources and access for global information about various objects and services.

Summary of Layers



TCP/IP PROTOCOL SUITE

The layers in the TCP/IP protocol suite do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport, and application. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: physical, data link, network, transport, and application.

The first four layers provide physical standards, network interfaces, internetworking, and transport functions that correspond to the first four layers of the OSI model.

The three topmost layers in the OSI model, however, are represented in TCP/IP by a single layer called the *application layer*.

At the *physical and data link layers*, TCP/IP does not define any specific protocol. It supports all standard protocols.

At the *network layer* (the internetwork layer), TCP/IP supports the Internetworking Protocol (IP) which in turn uses four supporting protocols: ARP, RARP, ICMP and IGMP

At the transport layer, TCP/IP defines three protocols: Transmission Control Protocol (TCP), User Datagram Protocol (UDP), and Stream Control Transmission Protocol (SCTP). At the network layer, the main protocol defined by TCP/IP is the Internetworking Protocol (IP); there are also some other protocols that support data movement in this layer.

Internetworking Protocol (IP)

A **best effort delivery service**, the term best effort means that IP provides no error checking or tracking. Datagram can travel along different routes and can arrive out of sequence or be duplicated. IP does not keep track of the routes and has no facility for reordering data grams once they arrive at their destination.

Address Resolution Protocol

The **Address Resolution Protocol (ARP)** is used to associate a logical address with a physical address.

Reverse Address Resolution Protocol

It is used when a computer is connected to a network for the first time or when a diskless computer is booted.

Internet Control Message Protocol

The **Internet Control Message Protocol (ICMP)** is a mechanism used by hosts and gateways to send notification of datagram problems back to the sender.

Transport Layer

IP is a **host-to-host protocol**, meaning that it can deliver a packet from one physical device to another. UDP and TCP are **transport level protocols** responsible for delivery of a message from a process (running program) to another process. A new transport layer protocol, SCTP, has been devised to meet the needs of some newer applications.

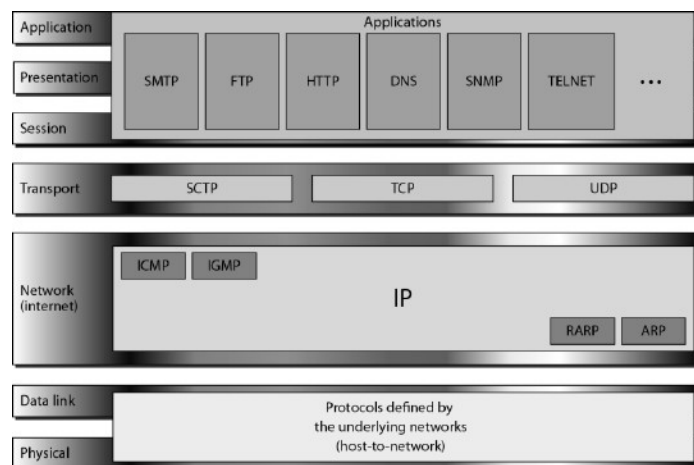
Transmission Control Protocol

TCP is a reliable stream transport protocol. The term stream, in this context, means connection-oriented: A connection must be established between both ends of a transmission before either can transmit data.

At the sending end of each transmission, TCP divides a stream of data into smaller units called segments. Each segment includes a sequence number for reordering after receipt, together with an acknowledgement number for the segments received.

Segments are carried across the internet inside of IP datagram's. At the receiving end, TCP collects each datagram as it comes in and reorders the transmission based on sequence numbers.

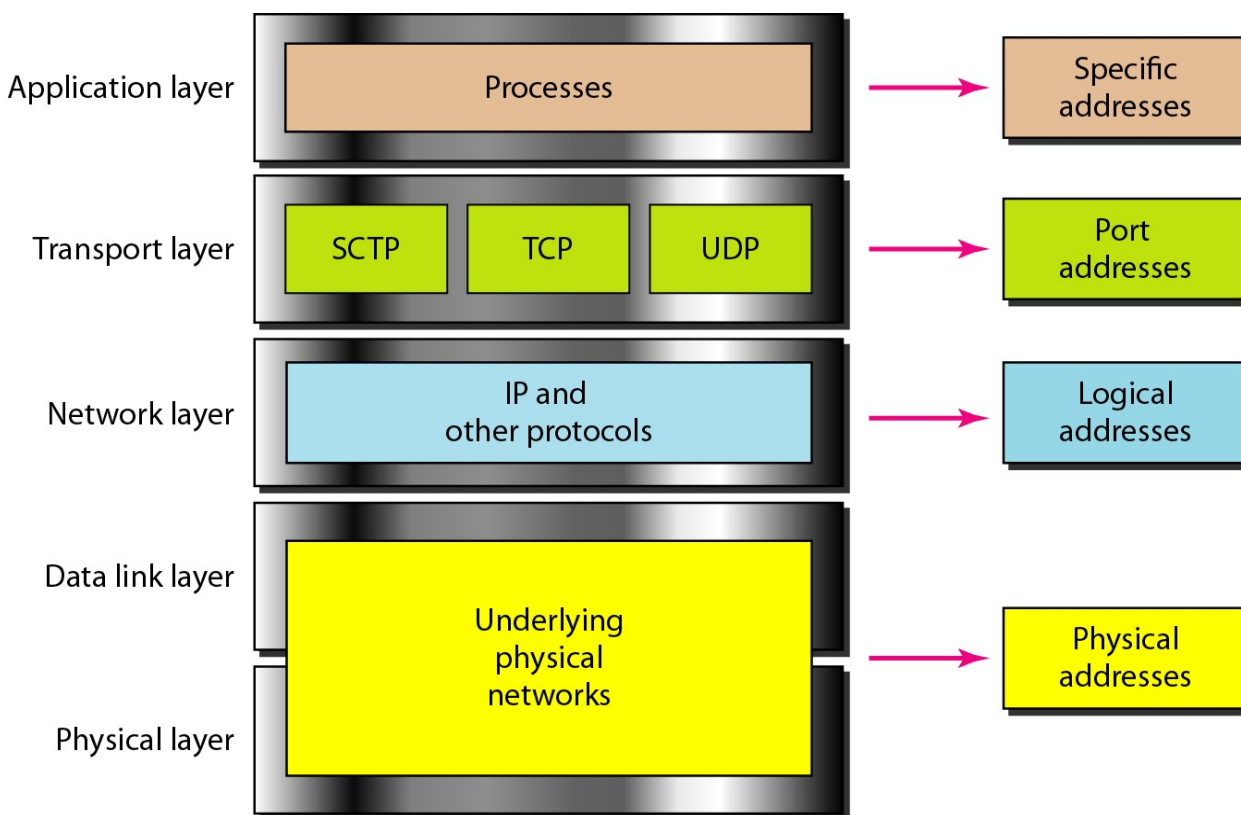
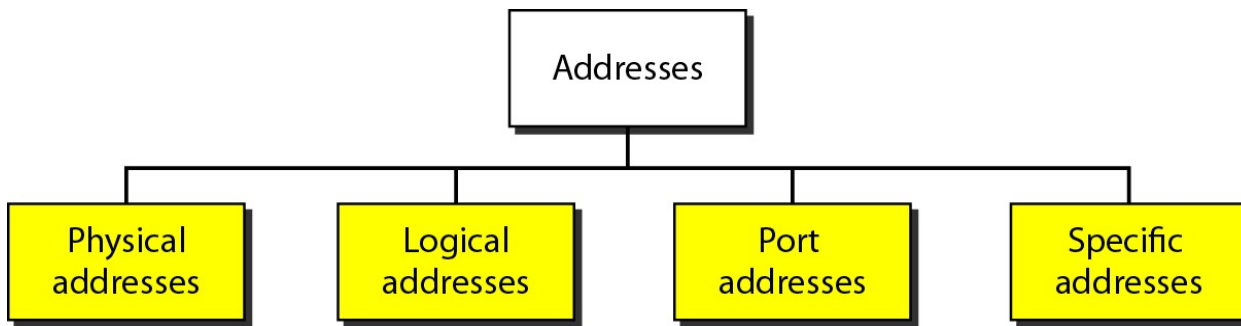
TCP/IP and OSI model



The application layer in TCP/IP is equivalent to session, presentation and application layers in the OSI Model.

Addressing:

Four levels of addresses are used in an internet employing the TCP/IP protocols: physical, logical, port, and specific



Most local-area networks use a 48-bit (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a colon, as shown below:

07:01:02:01:2C:4B

A 6-byte (12 hexadecimal digits) physical address

A port address is a 16-bit address represented by one decimal number as shown.

753

A 16-bit port address represented as one single number.

Transmission Media

- Concept of Guided and Unguided Transmission Media.
- Types of Guided Media.
- Types of Unguided Media.

3.1 Introduction - Need of Transmission Media, Selection Criteria.

3.2 Types of Transmission Media-

1. **Guided Media:** Cable Characteristics, Types of Cable-Twisted Pair Cable, Co-axial Cable, Fiber Optic Cable.
2. **Unguided media:** Types of Communication Band, Microwave Communication, Radio-wave Communication, Satellite Communication, Infrared Communication.

3.3 Latest Technologies in Wireless Network-Bluetooth Architecture, Wi-Fi, Wi- Max.

3.4 Cellular (Mobile) Telephone - Band in Cellular Telephony, Calls using Mobile Phones, Transmitting/Receiving / Handoff operations.

Introduction

In a data transmission system, the **transmission medium** is the physical path between transmitter and receiver.

Transmission media carries the information from sender to receiver. We use different types of cables or waves to transmit data. Data is transmitted normally through electrical or electromagnetic signals.

An electrical signal is in the form of current. An electromagnetic signal is series of electromagnetic energy pulses at various frequencies. These signals can be transmitted through copper wires, optical fibers, atmosphere, water and vacuum. Transmission media is also called **Communication channel**.

Criteria for selection of Transmission Media.

Different Medias have different properties like **bandwidth, delay, cost and ease of installation and maintenance**. The data transmission capabilities of various Media vary depending upon the various factors. These factors are:

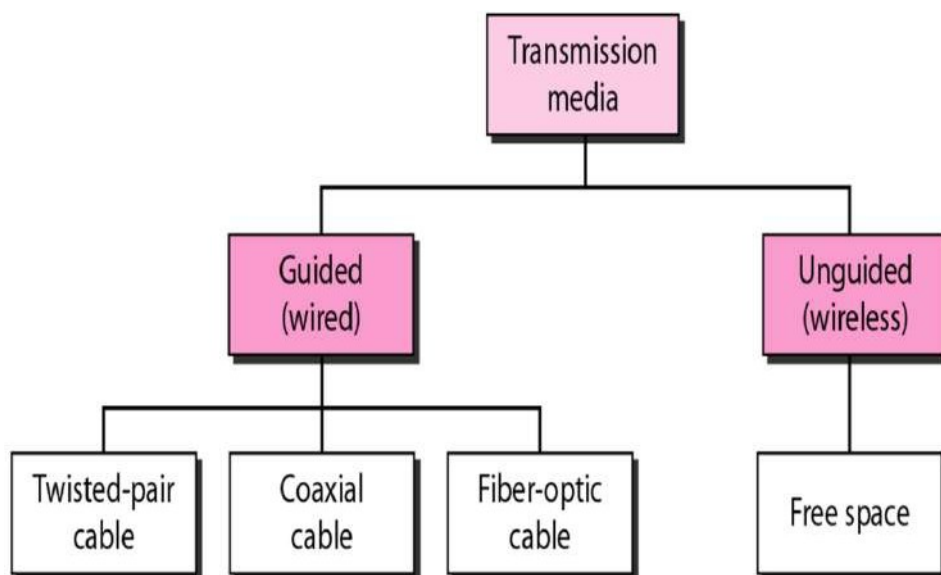
1. **Type of Media (Wired or Wireless).**

2. **Flexibility.** In order to expand network.
3. **Bandwidth.** It refers to the data carrying capacity of a channel or medium. Higher bandwidth communication channels support higher data rates.
4. **Reliability.** The consistency of transmission media (effect of weather conditions).
5. **Radiation.** It refers to the leakage of signal from the medium due to undesirable electrical characteristics of the medium.
6. **Noise Absorption.** It refers to the susceptibility of the media to external electrical noise that can cause distortion of data signal.
7. **Attenuation.** It refers to loss of energy as signal propagates outwards. The amount of energy lost depends on frequency.
8. **Number of receivers.** The number of users to be connected.
9. **Transmission Rate.**
10. **Cost and Ease of Installation.**
11. **Distances, etc.**

Types of Transmission Medias

There are two categories of transmission media used in computer communications.

1. **Guided Media (Wired)**
2. **Unguided Media (Wireless Media)**



LC 4: Physical Layer: Different Types of transmission media: Guided

1. Guided Media (Wired)

In a Guided transmission media the signals are sent through to a specific (solid) path using wire or cable. Guided media are made up of **copper conductor** bounded by **jacket (insulation)** material. Guided media is used high speed, good security and low cost requirements. Guided media is used in point to point communication.

Guided media are further divided in three Types.

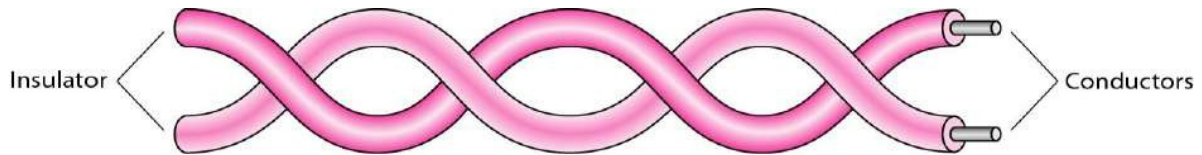
- 1. Twisted Pairs Cable**
- 2. Coaxial Cable**
- 3. Fiber Optics Cable**

1. Twisted Pair cable

- The **least expensive** and **most widely used guided transmission medium**.
- It is **lightweight, cheap, can be installed easily**, and they support many different types of network.
- A twisted pair consists of **two insulated copper wires** arranged in a regular **spiral pattern**.
- Typically, a number of these pairs are bundled together into a cable by wrapping them in a tough protective sheath.
- Over longer distances, cables may contain hundreds of pairs.
- Twisted pair (TP) maybe used to **transmit both analog and digital signal**.
- For analog signals amplifiers are required about every 5 to 6 kms. For digital signals repeaters are required every 2 to 3kms.

Why twist a cable?

- The twisting tends to **decrease the crosstalk (EMI) interference** between adjacent pairs in a cable.
- Neighboring pairs in a bundle typically have somewhat different twist lengths to reduce the crosstalk interference.
- On long-distance links, the twist length typically varies from 5 to 15



Application:-

1. It is the most commonly used medium of telephone n/w.
2. In the telephone system individual residential telephone sets are connected to the local telephone exchange or to end office by twisted pair wire.
3. Twisted pair also the most common medium used for the digital signaling for connections to a digital data switch.
4. It is also commonly used within a building for LAN.

There are two types of twisted pair:-

- A. Unshielded twisted pair (UTP).
- B. Shielded twisted pair (STP).



Fig. UTP and STP cables

A. Unshielded Twisted Pair (UTP):-

- Usually consists of two copper wires wrapped in individual plastic insulation.
- UTP cables are the most common telecommunications medium.
- The frequency range of the twisted pair cables enable both voice and data transmission.
- UTP cables consist of 2 or 4 pairs of twisted cable. Cable with 2 pair use **RJ- 11** connector and 4 pair cable use **RJ-45** connector.



(a)

There are five categories of UTPs:

- **Category 1:-** These originally used only for voice communication and can support only low data rates.
- **Category 2:-** Suitable for Voice and Data gives speed upto 4 mbps. This can't be used for high speed data communication. Older n/w's use this category.

- **Category 3:-** It is suitable for most PC n/w's support data rate of up to 16mbps currently most telephone n/w uses this.
- **Category 4:-** It offers data rate up to 20mbps.
- **Category 5:-**It offers data rate of 100mbps. Can be used for fast **Ethernet**. It requires more insulation and more twist per foot. It requires compatible equipment's.

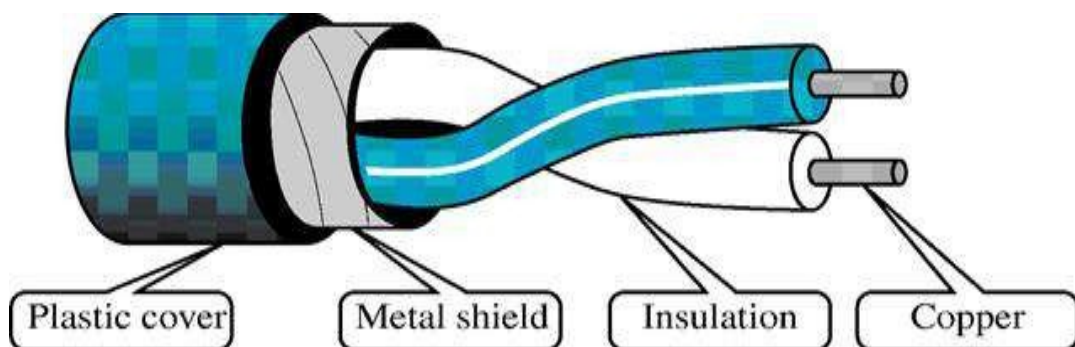
Advantages:

- Installation is easy
- Flexible
- Cheap
- It has high speed capacity,
- 100-meter limit
- Higher grades of UTP are used in LAN technologies like Ethernet.

Disadvantages:

- Bandwidth is low when compared with Coaxial Cable
- Provides less protection from interference (EMI).

2. Shielded twisted pair (STP).



- The only difference between STP and UTP is that STP cables have a shielding in usually of aluminum or polyester material between the outer jacket and wire.
- The shield makes STP less vulnerable to EMI, because the shield is electrically grounded.
- The metal mesh around the insulated wires eliminates **crosstalk**.
- **Crosstalk** occurs when one line picks up some of the other signals traveling down another line.

Advantages:

- Easy to install
- Performance is adequate
- Can be used for Analog or Digital transmission

- Increases the signaling rate
- Higher capacity than unshielded twisted pair
- Eliminates crosstalk

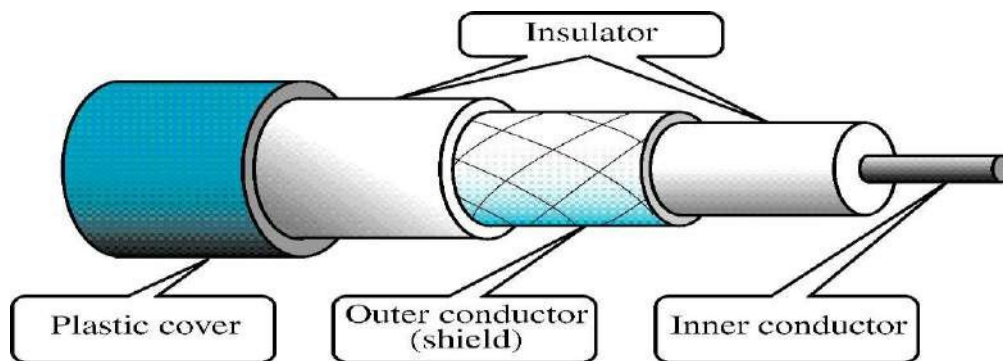
Disadvantages:

- Difficult to manufacture
- Heavy
- Expensive than UTP

Difference UTP vs STP

Factors	UTP	STP
Bandwidth	1-155Mbps	1-155Mbps
Node capacity per segment	2	2
Attenuation	High	Low
EMI	Very High	High
Installation	Easy	Fairly Easy
Cost	Low	Moderate

2. Co-axial cable



- The name coaxial is because it contains two conductors that are parallel to each other and share common axis.
- Inner conductor is made of copper which is surrounded by PVC insulation.
- The outer conductor is metal foil, mesh or both.
- Outer metallic conductor is used as a shield against noise.
- The outer conductor is also encased in an insulating sheath.
- The outermost part is the plastic cover which protects the whole cable.
- Co-axial cable is much less susceptible to interference and cross talk than the twisted pair.
- Co-axial cable is used to transmit both analog and digital signal.

Application –

Co-axial cable is widely used in the wide variety of applications. The most important of these are: -

1. TV distribution: -

Co-axial cable is spreading rapidly as a means of distributing TV signals to individual homes –cables TV. A cable TV system can carry dozens or even hundreds of TV channels at ranges up to a few tens of miles.

2. Long distance telephone transmission:-

Co-axial cable has traditionally been as important part of the long distance telephone n/w using FDM (frequency division multiplication) a co-axial cable can carry over 10,000 voice channels simultaneously.

3. Short distance communication links: -

Co-axial cable is also commonly used for short range connection between devices.

E.g. can be used to provide high speed I/O channels for an PC system.

4. LAN: -

Co-axial cable can be support a large number of devices with a variety of data and traffic types over distance that covers single building or a complex of building.

Co-axial cables usually for data transmission are of two types.

- 10 base 2 thin net - Node capacity per segment for 10 base 2 is 30.
- 10 base 2 thick net- Node capacity for 10 base 5 is 100.

There are two types of Coaxial cables:

BaseBand

This is a 50 ohm (Ω) coaxial cable which is used for digital transmission. It is mostly used for LAN's. Baseband transmits a single signal at a time with very high speed. The major drawback is that it needs amplification after every 1000 feet.

BroadBand

This uses analog transmission on standard cable television cabling. It transmits several simultaneous signal using different frequencies. It covers large area when compared with Baseband Coaxial Cable.

Advantages:

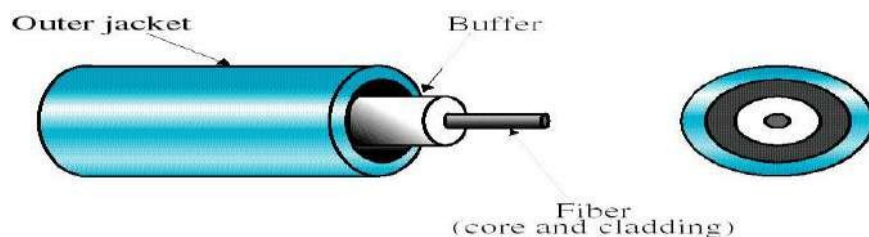
- Bandwidth is high
- Used in long distance telephone lines.
- Transmits digital signals at a very high rate of 10Mbps.
- Much higher noise immunity
- Data transmission without distortion.
- The can span to longer distance at higher speeds as they have better shielding when compared to twisted pair cable

Disadvantages :

- Single cable failure can fail the entire network.
- Difficult to install and expensive when compared with twisted pair.
- If the shield is imperfect, it can lead to grounded loop.

3. Optical Fiber or fiber optics

- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.
- A light pulse can be used to signal a one (1) bit.
- The absence of a pulse signals a zero(0).
- The bandwidth of an optical transmission system is potentially enormous.



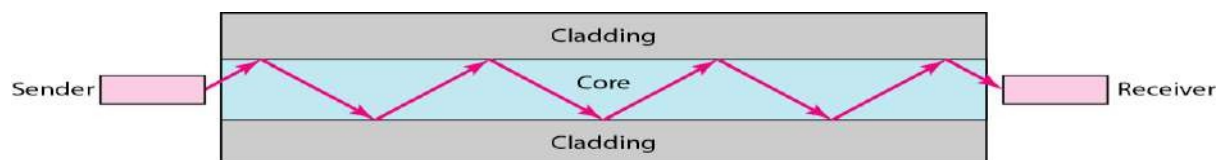
An optical fiber has an cylindrical shape and consists of 3 concentric section –

- (i) Core
- (ii) Cladding
- (iii) Jacket

1. **Core:-** It's the inner most section is made of glass or plastic and is surrounded by its own cladding. The core diameter is in the range of **8 to 50 μm** .
2. **Cladding:-** A glass or plastic coating that has optical properties different from those of the core having a diameter of **125 μm** . The cladding acts as a **reflector to light** that would otherwise escape the core.
3. **Jacket:-** The outer most layers surrounding caddied fiber is the jacket. Jacket is composed of plastic or other material layer to protect against moisture, cut, crushing and other environmental dangers.

Optical Fiber Communication

- A transmitter (Light Source) at senders end sends a Light across the fiber.
- A receiver at the other end makes use of Light Sensitive transistor to detect the absence or presence of light to indicate 0 or 1.



- The transmission medium is an ultra-thin fiber of glass.
- Light enters the cylindrical glass or plastic core at small angles are reflected and propagates along the fiber.

- The detector generates an electrical pulse when light falls on it.

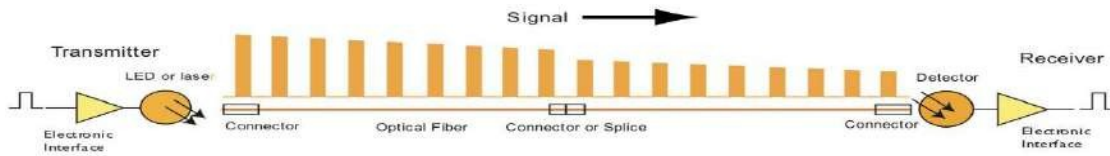


Fig. OFC Communication

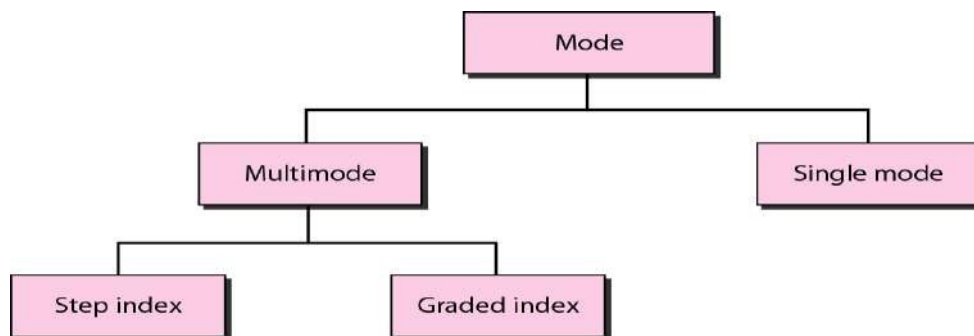
Two different types of **light sources** are used in fiber optic system.

- The Light Emitting Diode (LED)
- Injection Laser Diode (ILD)

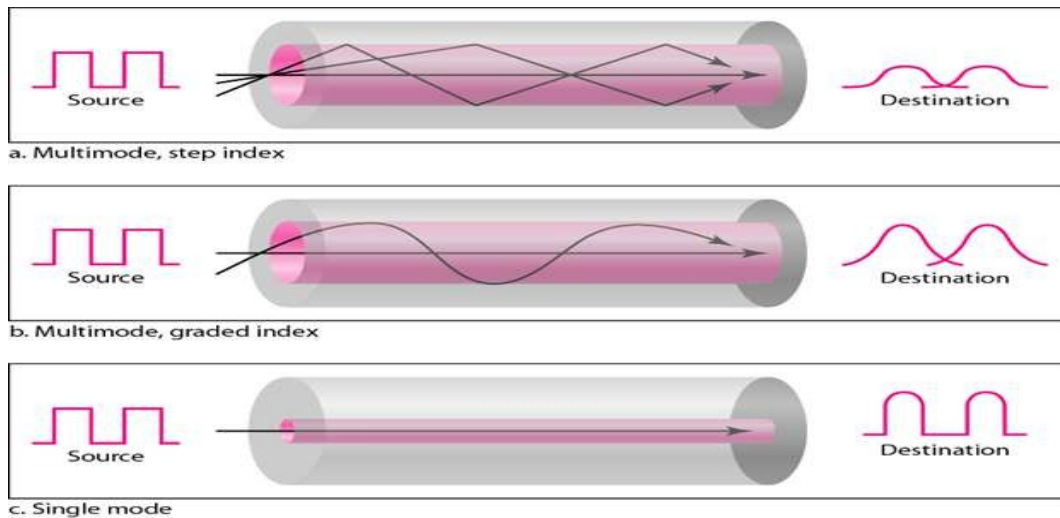
Both are semiconductor devices that emit a beam of light when voltage is applied. Led is less costly than ILD. ILD operates on laser principle, is more efficient, and can sustain greater data rate.

Types of Fiber Propagation Modes

- Optical fiber may be multi-mode or single mode.
- Single mode fibers allow a single light pass and are used with laser signaling. Single mode fibers can allow greater bandwidth and cable runs than multimode, but it is more expensive.
- Multimode fibers use multiple light pass the physical characteristics of the multiple mode fiber make all parts of the signal arrive at the same time appearing to the receiver as though they were one pulse.



1. Multimode **step-index fiber**
 - the reflective walls of the fiber move the light pulses to the receiver
2. Multimode **graded-index fiber**
 - acts to refract the light toward the center of the fiber by variations in the density
3. **Single mode fiber**
 - the light is guided down the center of an extremely narrow core



Advantages :

- Provides high quality transmission of signals at very high speed (bandwidth 2 Gbps)
- These are not affected by electromagnetic interference, so noise and distortion is very less.
- **Highly secure** due to tap difficulty and lack of signal radiation.
- Used for both analog and digital signals.
- **Smaller size and light weight**
- **Lower attenuation**

Disadvantages :

- It is expensive
- Difficult to install. requires **highly skilled installers**
- Maintenance is expensive and difficult.
- Do not allow complete routing of light signals.

Applications

- ◆ Telephones, including cellular wireless ◆
- Internet
- ◆ LANs - local area networks
- ◆ CATV - for video, voice and Internet connections ◆
- Utilities - management of power grid
- ◆ Security - closed-circuit TV and intrusion sensors ◆
- Transportation – smart lights and highways
- ◆ Military – everywhere!

Characteristics of cable media:-

Factor	UTP	STP	Co-axial	Fiber optics
Cost	Low	Moderate	Moderate	Highest

Installation	Easy	Fairly easy	Fairly easy	Difficult
Data rate	1 to 155 mbps	1 to 155 mbps	500 mbps	2 GBPS
Node capacity	2	2	30-100	2
Attenuation	High(100's of meter)	High(100's of meter)	Lower (range of few km's)	Lowest (10 Km's)
EMI	Most vulnerable	Less vulnerable than UTP	Less vulnerable than UTP	Not effected by EMI
Bandwidth	Low	Moderate	Moderatly high	Very high
Signals	Electrical	Electrical	Electrical	Light

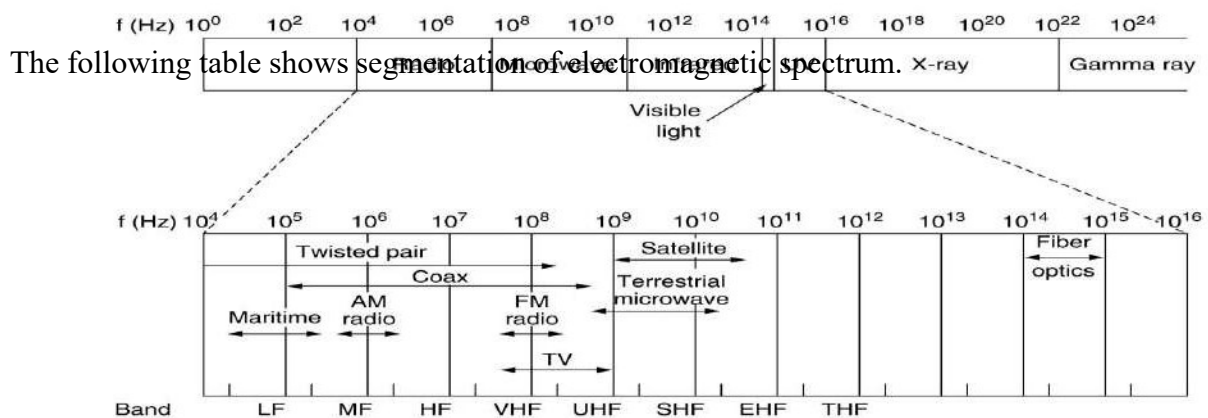
LC 5: Physical Layer: Different Types of transmission media: Unguided

Unguided Media

- Unguided media transmit electromagnetic waves without using a physical conductor.
- This is also called as wireless communication.
- Signals are normally broadcast through free space and thus are available to anyone who has **a device capable of receiving them.**

Electromagnetic Spectrum (Communication Band)

Electromagnetic spectrum is used for wireless communication. It is divided into various sub-bands.



<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3-30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30-300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz-3 MHz	Sky	AM radio
HF (high frequency)	3-30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30-300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz-3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3-30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30-300 GHz	Line-of-sight	Radar, satellite

Propagation Methods

Unguided signals can travel from the source to destination in several ways:

- Ground propagation,
- Sky-Propagation, and
- Line-of-Sight Propagation

1. Ground propagation mode:

- Radio waves **travel close to the earth**.
- These **low-frequency signals** proceed in all directions from the transmitting antenna and follow the **curvature of the planet**.
- **Distance** depends on the amount of **power in the signal**: The greater the power, the greater the distance.



2. Sky propagation mode:

- In this **high-frequency radio waves** radiate upward into the ionosphere where **they are reflected back** to earth.
- Ionosphere is the layer of atmosphere where **particles exist as ions**.
- This type of transmission allows for **greater distances with lower output power**.

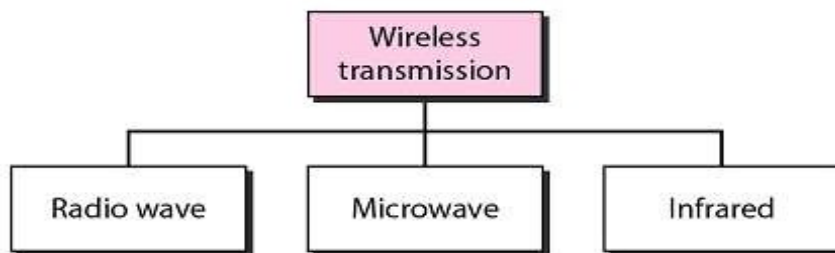


3. Line-of-sight propagation mode:

- In this, **very high-frequency** signals are transmitted **in straight lines** directly from **antenna to antenna**.
- Antennas must be **directional**, facing each other and tall enough.
- Line-of-sight propagation **is tricky** because radio transmissions cannot be completely focused.



Classification of Wireless Media



- Wireless communication may be via:
 - **Radio frequency** communication
 - **Microwave** communication
 - **Infrared** short range communication

1. Radio Wave Transmission

- Electromagnetic waves ranging in frequency between **3 kHz and 1 GHz** are normally called radio waves.
- Radio waves, are **omnidirectional**, i.e. they are **propagated in all directions**.
- A sending antenna sends waves that can be received by **any receiving antenna**.



- Radio waves, particularly of **low and medium frequencies**, can penetrate walls.
- It is an **advantage** because, an **AM radio** can receive signals **inside a building**.
- It is a **disadvantage** because we **cannot isolate** a communication to just **inside or outside a building**.

Applications

- The omnidirectional characteristics of radio waves make them useful for **multicasting**, in which there is **one sender but many receivers**.
- **AM and FM radio**,
- **Television**,
- **Maritime radio**,
- **Cordless phones and paging**.

2. Microwave Transmission

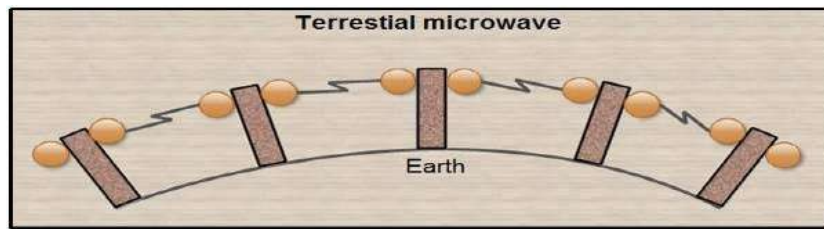
- Electromagnetic waves having frequencies between **1GHz and 300 GHz** are called microwaves.
- Microwaves are **unidirectional**.
- When an antenna transmits microwave waves, they can be **narrowly focused**.
- This means that the **sending and receiving antennas need to be aligned**.
- The unidirectional property has an obvious advantage.
 - A pair of antennas can be aligned without interfering with another pair of aligned antennas.
- **Microwaves use line-of-sight transmission**.
- This means that microwaves must be transmitted in straight line and no obstructions, such as buildings or mountains, between microwave stations.
- To avoid possible obstructions, microwave antennas often are positioned on the **tops of buildings, towers, or mountains**.

Microwave transmission is divided into two types

A. Terrestrial Microwave

- Used for **long-distance telephone service**.
- Uses radio frequency spectrum, from **2 to 40 GHz**.
- Parabolic dish transmitter, mounted high.
- Requires **unobstructed line of sight** between source and receiver

- Curvature of the earth requires stations (**repeaters**) 30 miles apart



Advantages:

- Effect of noise is reduced because of repeaters.
- Maintenance is less as compared to cable.
- No interference with other transmission channels.

Disadvantages:

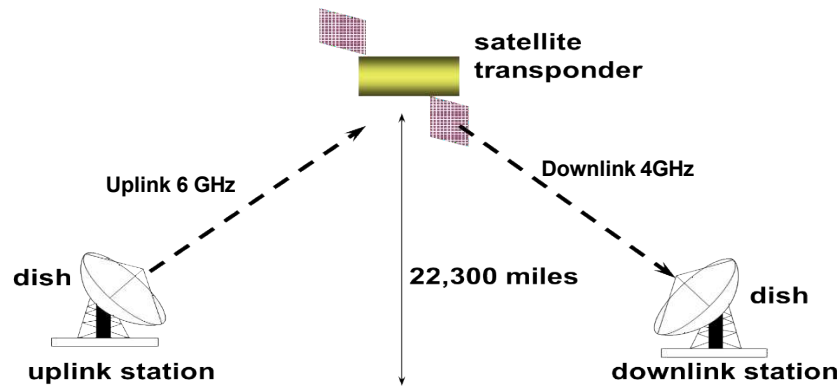
- Communication can be affected because of atmospheric phenomenon and passing airplanes and rain
- Line of sight requirement
- Expensive towers and repeaters.

Application:-

- Long-distance telecommunication service
 - requires fewer amplifiers or repeaters than coaxial cable
 - Example
 - telephone system
 - TV distribution
- Short point-to-point links
 - Data link between local area network
 - Closed-Circuit TV
- **By passing n/w:-** Microwave can also be used for bypass application. A business can establish a microwave link to long a distance telecommunication facility in the same city, by passing the local telephone company.

B. Satellite microwave:-

- A communication satellite acts like a microwave station.
- It is used to link two or more ground waves microwave transmitter or receiver known as earth stations.
- The satellite receives transmission on one frequency band (uplink), amplifies or repeats the signal and transmit it on another frequency (down link).
- A single orbiting satellite will operate on no. of frequency bands called transponder channels or simply transponder.



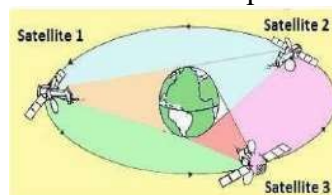
- In the fig the satellite is being used to provide a point to point link between two distant grounds based antenna.
- The signal, a beam of **modulated microwaves** is sent towards the satellite called **UPLINK (6 Ghz)**.
- Then the satellite processes the signal and send it back to the receiver's antenna present on the earth's surface called as **DOWNLINK (4Ghz)**.
- The satellite has to **receive, process** and **transmit** the signal.
- A unit called as Satellite Transponder performs all these functions.
- The communication satellite has two sets of transponders.
- Each set having **12 transponders**.
- Each transponder has a bandwidth of 36MHz.

Types of Satellite by there purpose

- Communication Satellite
- Weather satellite
- Remote- Sensing Satellite
- Scientific Satellite

Geostationary Earth Orbit (GEO)

- These satellites are in orbit 35,863 km above the earth's surface along the equator.
- Objects in Geostationary orbit revolve around the earth at the same speed as the earth rotates.
- This means GEO satellites remain in the same position relative to the surface of earth.



- **Principal Satellite Transmission Bands**
- **C band:**
 - 4(downlink) - 6(uplink) GHz
 - the first to be designated
- **Ku band:**
 - 12(downlink) -14(uplink) GHz
 - rain interference is the major problem
- **Ka band:**
 - 19(downlink) - 29(uplink) GHz
 - equipment needed to use the band is still very expensive

Application:-

- **Television** distribution
 - A network provides programming from a central location
 - Direct broadcast satellite (DBS)
 - Long-distance **telephone** transmission
 - High-usage international trunks
- Private business networks
- **Military** Applications
- Other applications
 - **digital** cinema
 - Satellite **radio**
 - Satellite **internet access**

3. Infrared

- Infrared waves, with frequencies from **300 GHz to 400 THz** (wavelengths from **1 mm to 770 nm**),
- Used for **short-range communication**.
- Infrared communication is achieved using transmitters/receivers (Transceivers) that modulate non-coherent infrared light.
- Transceiver must be in line of sight of each other either directly or via reflection from light colored surface such as the ceiling of the room.
- One important difference between infrared and microwave transmission is that they don't penetrate walls.
- The **remote controls** used for televisions, VCRs, and stereos all use infrared communication.
- They are relatively directional, cheap, and easy to build.

Applications

- TV Remote control
- Guidance in weapon system
- Wireless keyboards and mouse.

LC 6: Multiplexing: TDM, FDM and WDM:

Multiplexing – Types of multiplexing – LAN – Project 802 – Ethernet – Token bus – Token ring – FDDI – MAN – IEEE 802.6 – Circuit switching – Packet switching

1.1 MULTIPLEXING

Multiplexing is the name given to techniques, which allow more than one message to be transferred via the same communication channel. The channel in this context could be a transmission line, *e.g.* a twisted pair or co-axial cable, a radio system or a fibre optic system *etc.*

A channel will offer a specified bandwidth, which is available for a time t , where t may ∞ or 0 . Thus, with reference to the channel there are 2 'degrees of freedom', *i.e.* bandwidth or frequency and time.

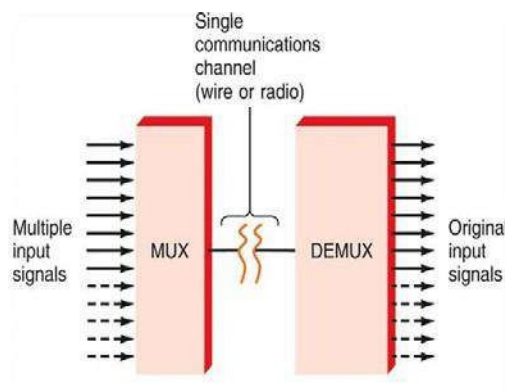
Now consider a signal $v_s(t) = A \cos(\omega t + \phi)$. The signal is characterised by amplitude, frequency, phase and time.

Various multiplexing methods are possible in terms of the channel bandwidth and time, and the signal, in particular the frequency, phase or time.

Concept of Multiplexing

Multiplexing is a technique by which different analog and digital streams of transmission can be simultaneously processed over a shared link. Multiplexing divides the high capacity medium into low capacity logical medium which is then shared by different streams.

- ☐ Communication is possible over the air (radio frequency), using a physical media (cable) and light (optical fiber). All mediums are capable of multiplexing.
- ☐ When more than one sender tries to send over single medium, a device called



Multiplexer divides the physical channel and allocates one to each. On the other end of communication, a De-multiplexer receives data from a single medium and identifies each and send to different receivers.

- Transmitting two or more signals simultaneously can be accomplished by running multiple cables or setting up one transmitter receiver pair for each channel , but this is an expensive approach.

A single cable or radio link can handle multiple signals simultaneously using a technique known as multiplexing. Multiplexing permits hundreds or even thousands of signals to be combined and transmitted over a single medium.

- A device called a multiplexer (often shortened to "mux") combines the input signals into one signal. When the multiplexed signal needs to be separated into its component signals (for example, when your email is to be delivered to its destination), a device called a demultiplexer (or "demux") is used.

Multiplexing was originally developed in the 1800s for telegraphy. Today, multiplexing is widely used in many telecommunications applications, including telephony, Internet communications, digital broadcasting and wireless telephony.

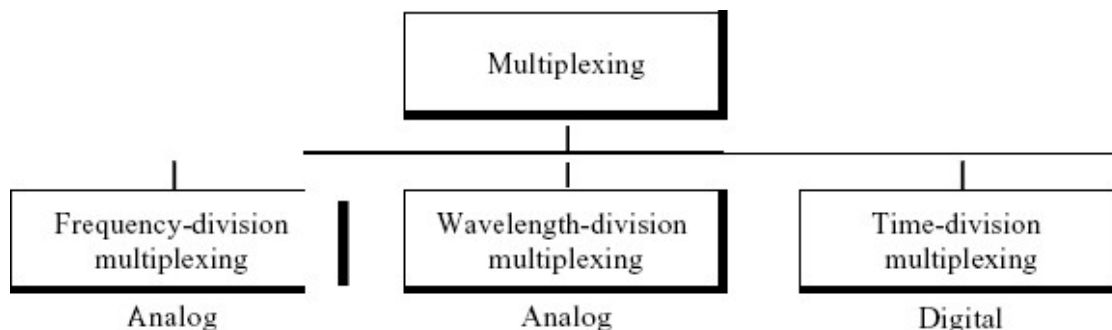


Figure *Categories of multiplexing*

1) Frequency Division Multiplexing FDM

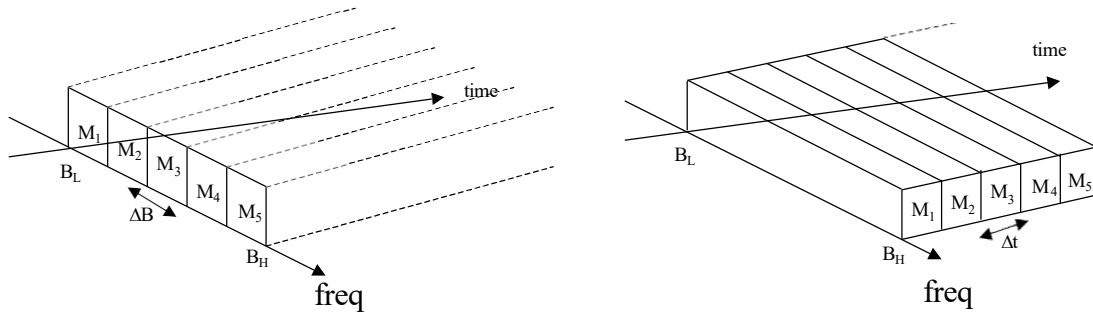
FDM is derived from AM techniques in which the signals occupy the same physical 'line' but in different frequency bands. Each signal occupies its own specific band of frequencies all the time, *i.e.* the messages share the channel **bandwidth**.

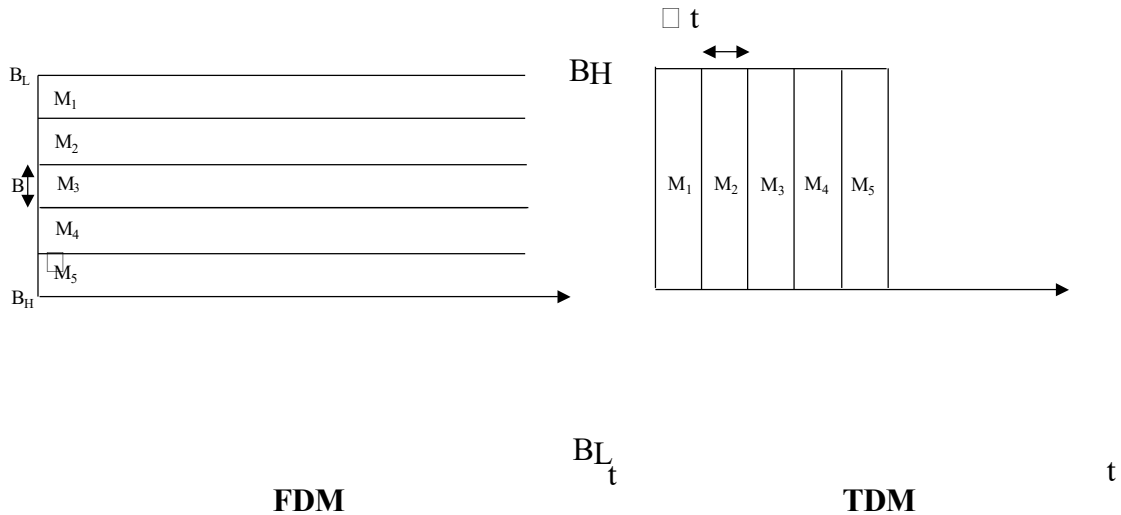
2) Time Division Multiplexing TDM

TDM is derived from sampling techniques in which messages occupy all the channel bandwidth but for short time intervals of time, *i.e.* the messages share the channel **time**.

- FDM – messages occupy **narrow** bandwidth – all the time.
- TDM – messages occupy **wide** bandwidth – for short intervals of time.

These two basic methods are illustrated below.



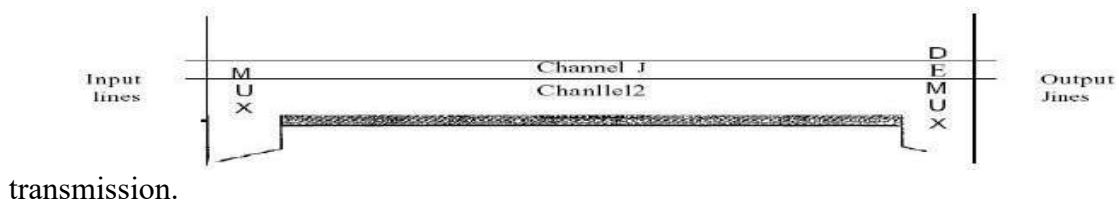


Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.

Frequency-Division Multiplexing

Frequency-division multiplexing (FDM) is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted. In FDM, signals generated by each sending device modulate different carrier frequencies. These modulated signals are then combined into a single composite signal that can be transported by the link. Carrier frequencies are separated by sufficient bandwidth to accommodate the modulated signal. These bandwidth ranges are the channels through which the various signals travel. Channels can be separated by strips of unused bandwidth-guard bands-to prevent signals from overlapping. In addition, carrier frequencies must not interfere with the original data frequencies.

Figure 1 gives a conceptual view of FDM. In this illustration, the transmission path is divided into three parts, each representing a channel that carries one



transmission.

Fig: FDM

We consider FDM to be an analog multiplexing technique; however, this does not mean that FDM cannot be used to combine sources sending digital signals. A digital signal can be converted to an analog signal before FDM is used to multiplex them.

Multiplexing Process

Figure 2 is a conceptual illustration of the multiplexing process. Each source generates a signal of a similar frequency range. Inside the multiplexer, these similar signals modulates different carrier frequencies (f_1, f_2 , and f_3). The resulting modulated signals are then combined into a single composite signal that is sent out over a media link that has enough bandwidth to accommodate it.

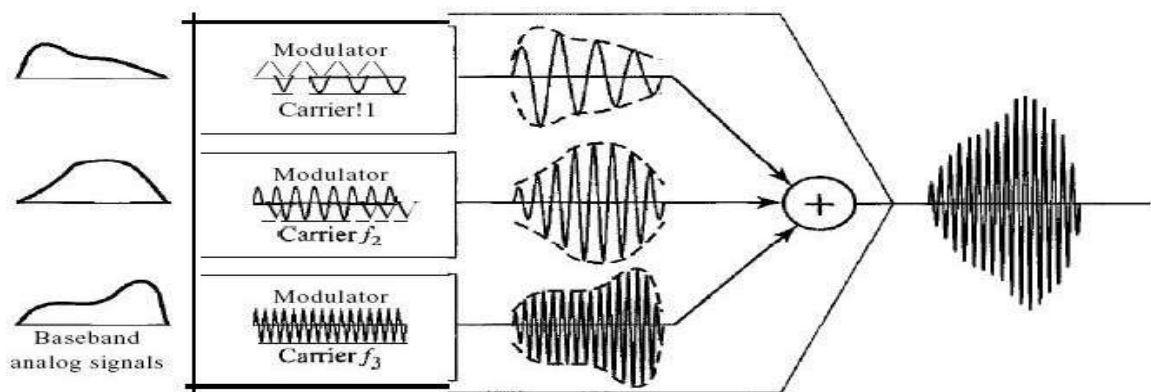


Figure 2 *FDM process*

Demultiplexing Process

The demultiplexer uses a series of filters to decompose the multiplexed signal into its constituent component signals. The individual signals are then passed to a demodulator that separates them from their carriers and passes them to the output lines. Figure 3 is a conceptual illustration of demultiplexing process.

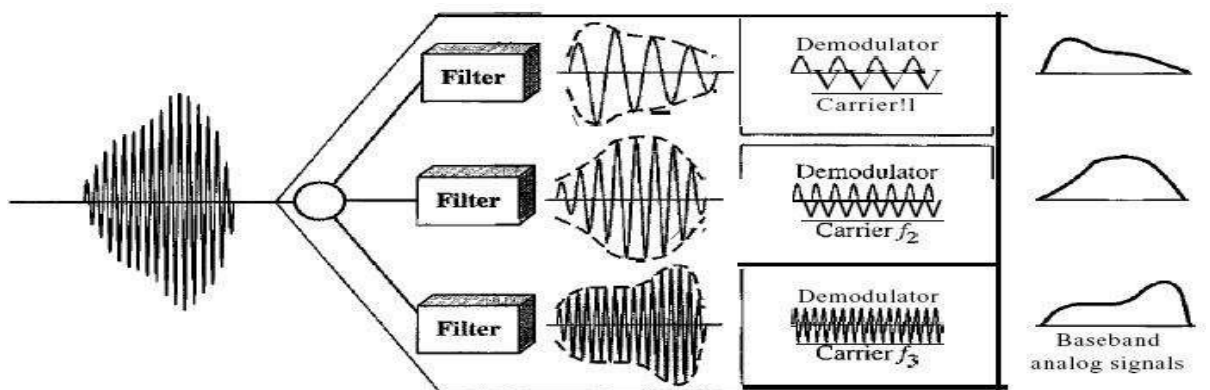
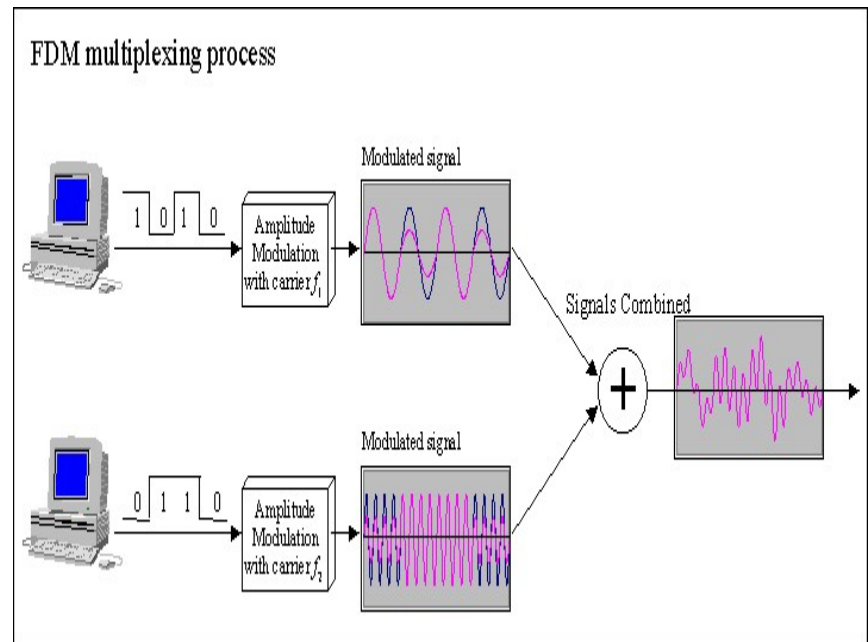


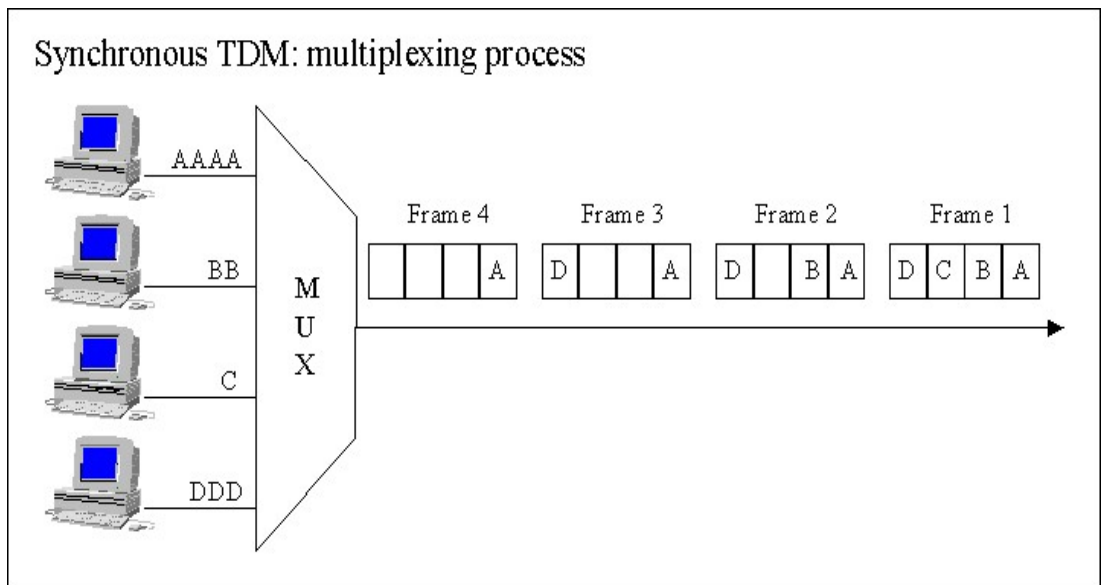
Figure 3 *FDM de-multiplexing example*



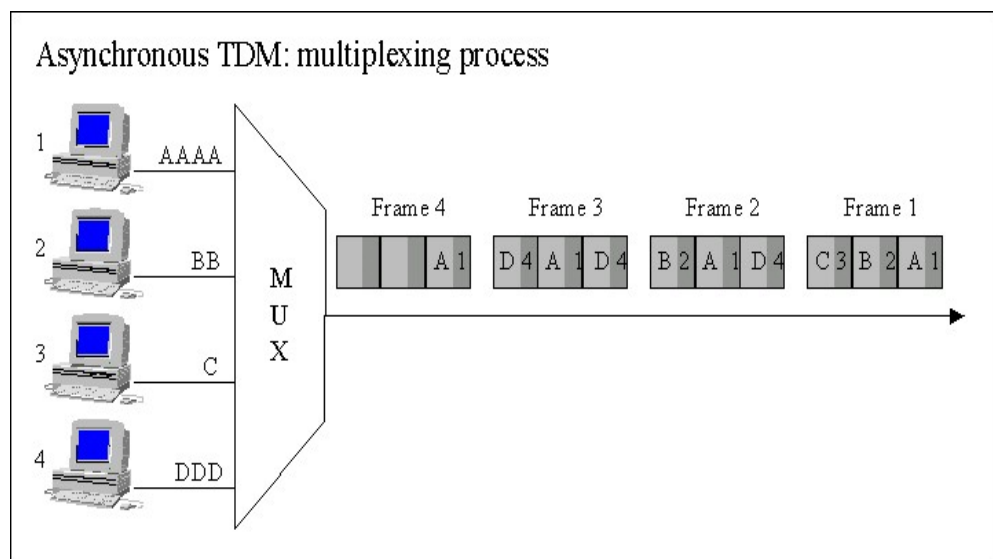
Time Division Multiplexing (TDM): This is possible when data transmission rate of the media is much higher than that of the data rate of the source. Multiple signals can be transmitted if

each signal is allowed to be transmitted for a definite amount of time. These time slots are so small that all transmissions appear to be in parallel.

1. **Synchronous TDM:** Time slots are preassigned and are fixed. Each source is given its time slot at every turn due to it. This turn may be once per cycle, or several turns per cycle, if it has a high data transfer rate, or may be once in a no. of cycles if it is slow. This slot is given even if the source is not ready with data. So this slot is transmitted empty.



2. **Asynchronous TDM:** In this method, slots are not fixed. They are allotted dynamically depending on speed of sources, and whether they are ready for transmission.



Wavelength-Division Multiplexing

Wavelength-division multiplexing (WDM) is designed to use the high-data-rate capability of fiber-optic cable. The optical fiber data rate is higher than the data rate of metallic transmission cable. Using a fiber-optic cable for one single line wastes the available bandwidth. Multiplexing allows us to combine several lines into one. WDM is conceptually the same as FDM, except that the multiplexing and demultiplexing involve optical signals transmitted through fiber-optic channels.

The idea is the same: We are combining different signals of different frequencies. The difference is that the frequencies are very high.

Figure 4 gives a conceptual view of a WDM multiplexer and demultiplexer. Very narrow bands of light from different sources are combined to make a wider band of light. At the receiver, the signals are separated by the demultiplexer.

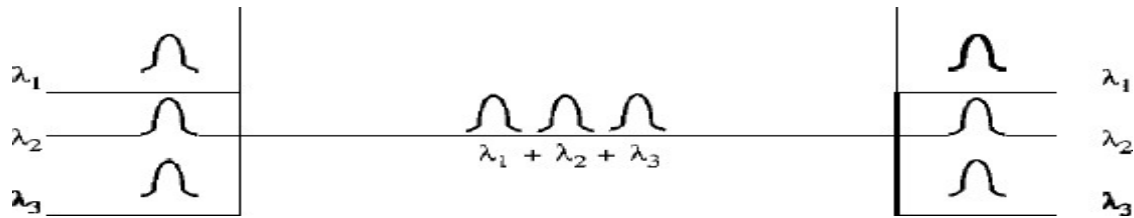


Figure 4 *Wavelength-division multiplexing*

Although WDM technology is very complex, the basic idea is very simple. We want to combine multiple light sources into one single light at the multiplexer and do the reverse at the demultiplexer. The combining and splitting of light sources are easily handled by a prism. Recall from basic physics that a prism bends a beam of light based on the angle of incidence and the frequency. Using this technique, a multiplexer can be made to combine several input beams of light, each containing a narrow band of frequencies, into one output beam of a wider band of frequencies. A demultiplexer can also be made to reverse the process. Figure 5 shows the concept.

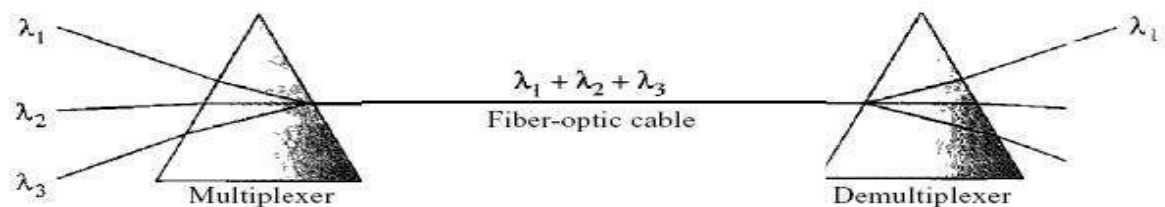


Figure 5 *Prisms in wavelength-division multiplexing and demultiplexing*

Synchronous Time-Division Multiplexing

Time-division multiplexing (TDM) is a digital process that allows several connections to share the high bandwidth of a line. Instead of sharing a portion of the bandwidth as in FDM, time is shared. Each connection occupies a portion of time in the link. Figure 6 gives a conceptual view of TDM. Note that the same link is used as in FDM; here, however, the link is shown sectioned by time rather than

by frequency. In the figure, portions of signals 1,2,3, and 4 occupy the link sequentially.

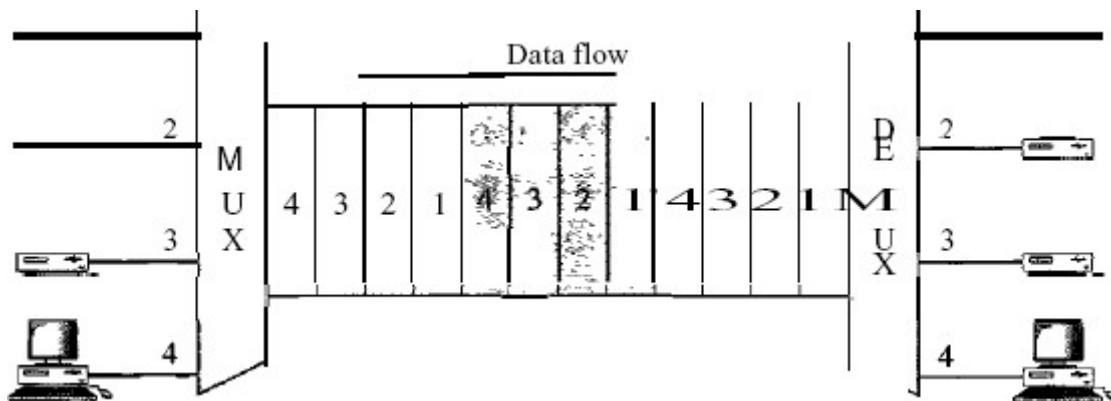


Figure 6 *TDM*

Note that in Figure 6 we are concerned with only multiplexing, not switching. This means that all the data in a message from source 1 always go to one specific destination, be it 1, 2, 3, or 4. The delivery is fixed and unvarying, unlike switching. We also need to remember that TDM is, in principle, a digital multiplexing technique. Digital data from different sources are combined into one timeshared link. However, this does not mean that the sources cannot produce analog data; analog data can be sampled, changed to digital data, and then multiplexed by using TDM.

We can divide TDM into two different schemes: synchronous and statistical.

In synchronous TDM, each input connection has an allotment in the output even if it is not sending data.

Time Slots and Frames

In synchronous TDM, the data flow of each input connection is divided into units, where each input occupies one input time slot. A unit can be 1 bit, one character, or one block of data. Each input unit becomes one output unit and occupies one output time slot. However, the duration of an output time slot is n times shorter than the duration of an input time slot. If an input time slot is T s, the output time slot is T/n s, where n is the number of connections. In other words, a unit

in the output connection has a shorter duration; it travels faster. Figure 7 shows an example of synchronous TDM where n is 3.

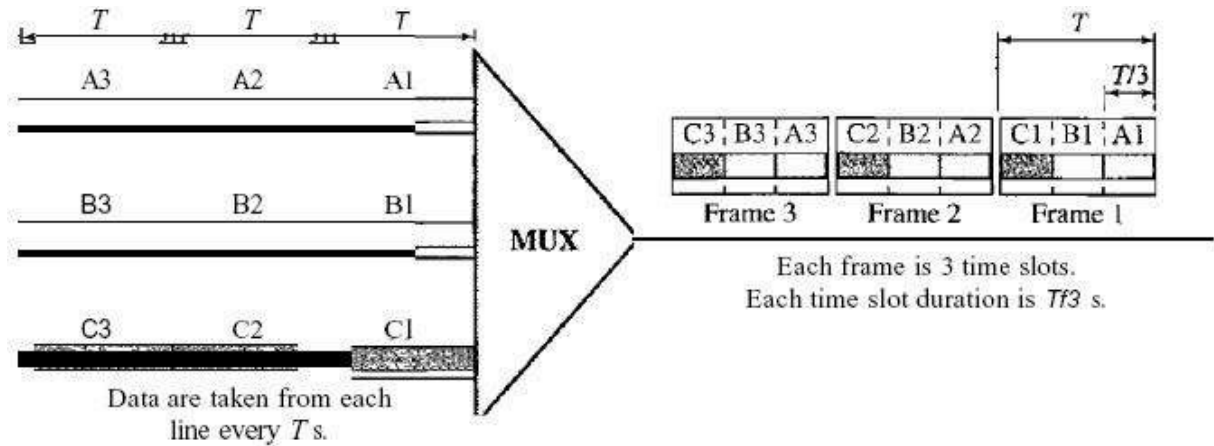


Figure 7 *Synchronous time-division multiplexing*

In synchronous TDM, a round of data units from each input connection is collected into a frame (we will see the reason for this shortly). If we have n connections, a frame is divided into n time slots and one slot is allocated for each unit, one for each input line. If the duration of the input unit is T , the duration of each slot is T/n and the duration of each frame is T (unless a frame carries some other information, as we will see shortly).

The data rate of the output link must be n times the data rate of a connection to guarantee the flow of data. In Figure 7, the data rate of the link is 3 times the data rate of a connection; likewise, the duration of a unit on a connection is 3 times that of the time slot (duration of a unit on the link). In the figure we represent the data prior to multiplexing as 3 times the size of the data after multiplexing. This is just to convey the idea that each unit is 3 times longer in duration before multiplexing than after. Time slots are grouped into frames. A frame consists of one complete cycle of time slots, with one slot dedicated to each sending device. In a system with n input lines, each frame has n slots, with each slot allocated to carrying data from a specific input line.

Interleaving

TDM can be visualized as two fast-rotating switches, one on the multiplexing side and the other on the demultiplexing side. The switches are synchronized and rotate at the same speed, but in opposite directions. On the multiplexing side, as the switch opens in front of a connection, that

connection has the opportunity to send a unit onto the path. This process is called **interleaving**. On the demultiplexing side, as the switch opens in front of a connection, that connection has the opportunity to receive a unit from the path.

Figure 8 shows the interleaving process for the connection shown in Figure 7.

In this figure, we assume that no switching is involved and that the data from the first connection at the multiplexer site go to the first connection at the demultiplexer

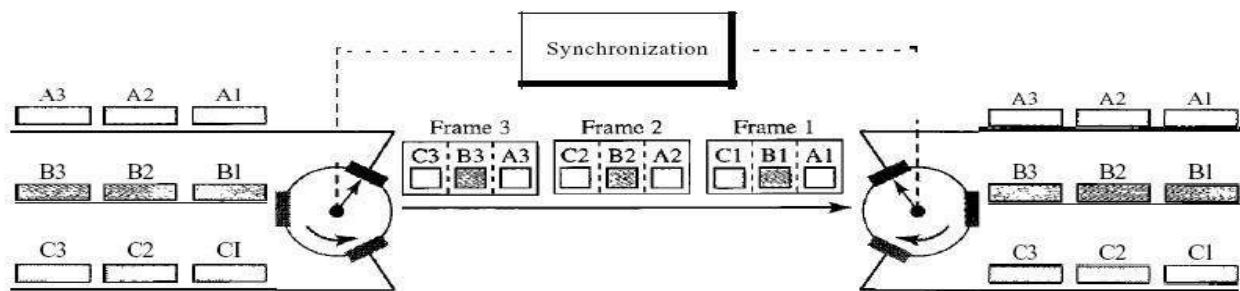


Figure 8 *Interleaving*

- *Empty Slots*
- *Data Rate Management*
 - *Multilevel Multiplexing*
 - **Multiple-Slot Allocation**
 - **P**

ulse

Stuffing •

Frame

Synchroniz

ing •

Digital

Signal

Service

Statistical Time-Division Multiplexing

Addressing

Figure a also shows a major difference between slots in synchronous TDM and statistical TDM. An output slot in synchronous TDM is totally occupied by data; in statistical TDM, a slot needs to carry data as well as the address of the destination.

In synchronous TDM, there is no need for addressing; synchronization and preassigned relationships between the inputs and outputs serve as an address. We know, for example, that input 1 always goes to input 2. If the multiplexer and the demultiplexer are synchronized, this is guaranteed. In statistical multiplexing, there is no fixed relationship between the inputs and outputs because there are no preassigned or reserved slots. We need to include the address of the receiver inside each slot to show where it is to be delivered. The addressing in its simplest form can be n bits to define N different output lines with $n = \log_2 N$. For example, for eight different output lines, we need a 3-bit address.

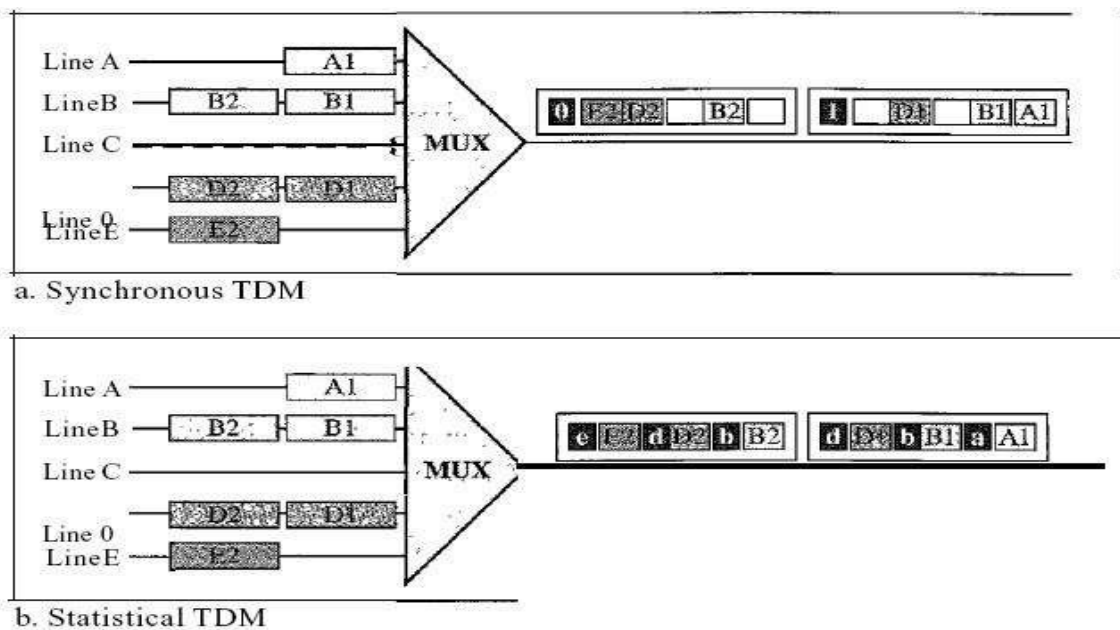


Figure a TDM slot comparison

Slot Size

Since a slot carries both data and an address in statistical TDM, the ratio of the data size to address size must be reasonable to make transmission efficient. For example, it would be inefficient to send 1 bit per slot as data when the address is 3 bits. This would mean an overhead of 300 percent. In statistical TDM, a block of data is usually many bytes while the address is just a few bytes.

No Synchronization Bit

There is another difference between synchronous and statistical TDM, but this time it is at the frame level. The frames in statistical TDM need not be synchronized, so we do not need synchronization bits.

Bandwidth

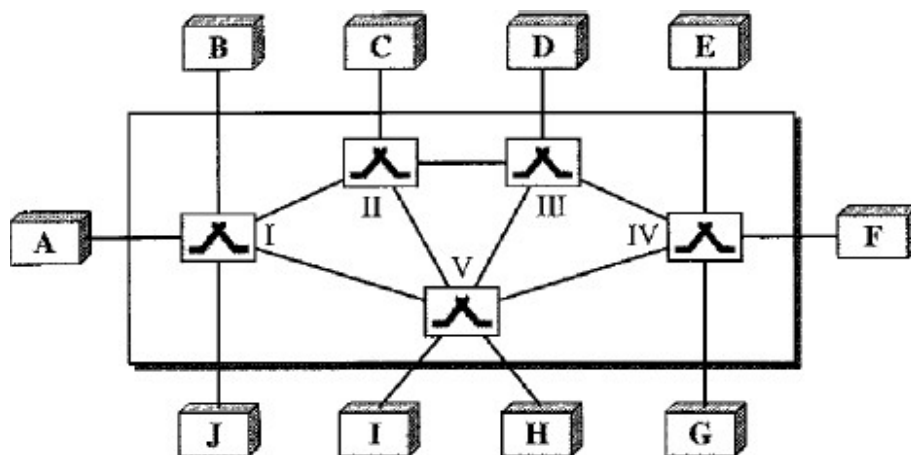
In statistical TDM, the capacity of the link is normally less than the sum of the capacities of each channel. The designers of statistical TDM define the capacity of the link based on the statistics of the load for each channel. If on average only x percent of the input slots are filled, the capacity of the link reflects this. Of course, during peak times, some slots need to wait.

LC 8:Switching and Taxonomy: Circuit Switched and Packet switched

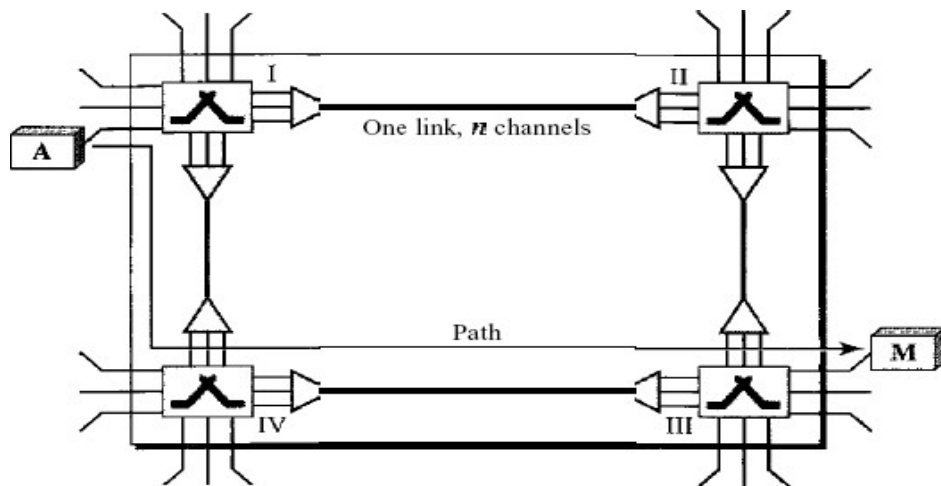
Switching

A network is a set of connected devices. Whenever we have multiple devices, we have the problem of how to connect them to make one-to-one communication possible. One solution is to make a point-to-point connection between each pair of devices (a mesh topology) or between a central device and every other device (a star topology). These methods, however, are impractical and wasteful when applied to very large networks. The number and length of the links require too much infrastructure to be cost-efficient, and the majority of those links would be idle most of the time. Other topologies employing multipoint connections, such as a bus, are ruled out because the distances between devices and the total number of devices increase beyond the capacities of the media and equipment.

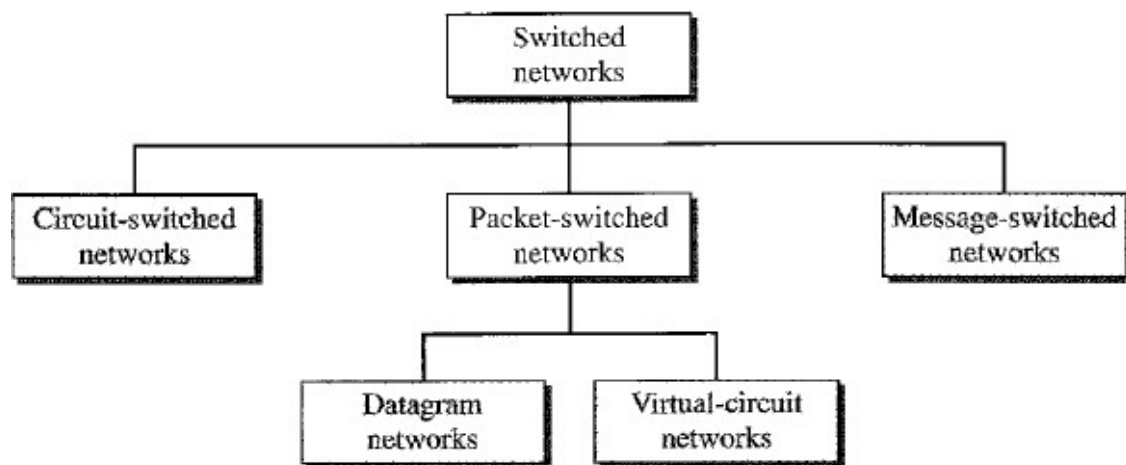
A better solution is switching. A switched network consists of a series of interlinked nodes, called switches. Switches are devices capable of creating temporary connections between two or more devices linked to the switch. In a switched network, some of these nodes are connected to the end systems (computers or telephones, for example). Others are used only



for routing. Figure shows a switched network.



The end systems (communicating devices) are labeled A, B, C, D, and so on, and the switches are labeled I, II, III, IV, and V. Each switch is connected to multiple links.



Taxonomy of switched networks

CIRCUIT-SWITCHED NETWORKS

A circuit-switched network consists of a set of switches connected by physical links. A connection between two stations is a dedicated path made of one or more links. However, each connection uses only one dedicated channel on each link. Each link is normally divided into n channels by using FDM or TDM

Figure shows a trivial circuit-switched network with four switches and four links. Each link is divided into n (n is 3 in the figure) channels by using FDM or TDM.

Three Phases

The actual communication in a circuit-switched network requires three phases: connection setup, data transfer, and connection teardown.

Setup Phase:

Before the two parties (or multiple parties in a conference call) can communicate, a dedicated circuit (combination of channels in links) needs to be established. The end systems are normally connected through dedicated lines to the switches, so connection setup means creating dedicated channels between the switches. For example, in Figure, when system A needs to connect to system M, it sends a setup request that includes the address of system M, to switch I. Switch I finds a channel between itself and switch IV that can be dedicated for this purpose. Switch I then sends the request to switch IV, which finds a dedicated channel between itself and switch III. Switch III informs system M of system A's intention at this time. In the next step to making a connection, an acknowledgment from system M needs to be sent in the opposite direction to system A. Only after system A receives this acknowledgment is the connection established. Note that end-to-end addressing is required for creating a connection between the two end systems. These can be, for example, the addresses of the computers assigned by the administrator in a TDM network, or telephone numbers in an FDM network.

Data Transfer Phase:

After the establishment of the dedicated circuit (channels), the two parties can transfer data.

Teardown Phase:

When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.

Efficiency:

It can be argued that circuit-switched networks are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection. These resources are unavailable to other connections. In a telephone network, people normally terminate the communication when they have finished their conversation. However, in computer networks, a computer can be connected to another computer even if there is no activity for a long time. In this case, allowing resources to be dedicated means that other connections are deprived.

Delay

Although a circuit-switched network normally has low efficiency, the delay in this type of network is minimal. During data transfer the data are not delayed at each switch; the resources are allocated for the duration of the connection. Figure 8.6 shows the idea of delay in a circuit-switched network when only two switches are involved. As Figure shows, there is no waiting time at each switch. The total delay is due to the time needed to create the connection, transfer data, and disconnect the circuit.

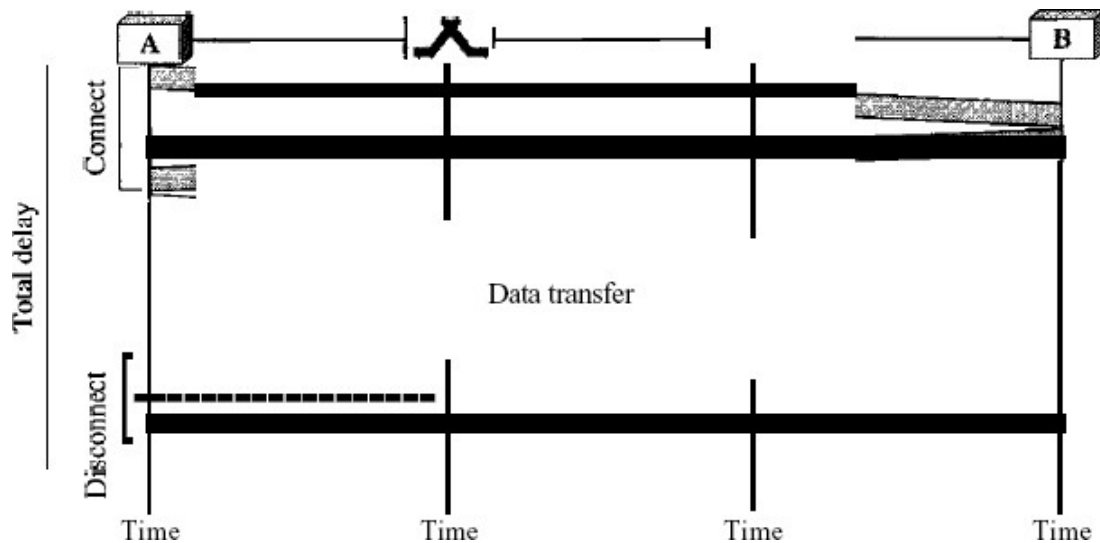


Fig: Delay in a circuit-switched network

The delay caused by the setup is the sum of four parts: the propagation time of the source computer request (slope of the first gray box), the request signal transfer time (height of the first gray box), the propagation time of the acknowledgment from the destination computer (slope of the second gray box), and the signal transfer time of the acknowledgment (height of the second gray box). The delay due to data transfer is the sum of two parts: the propagation time (slope of the colored box) and data transfer time (height of the colored box), which can be very long. The third box shows the time needed to tear down the circuit. We have shown the case in which the receiver requests disconnection, which creates the maximum delay.

DATAGRAM NETWORKS

In a datagram network, each packet is treated independently of all others. Even if a packet is part of a multipacket transmission, the network treats it as though it existed alone. Packets in this approach are referred to as datagrams.

Datagram switching is normally done at the network layer. We briefly discuss datagram networks here as a comparison with circuit-switched and virtual-circuit switched networks. Figure shows how the datagram approach is used to deliver four packets from station A to station

X. The switches in a datagram network are traditionally referred to as routers. That is why we use a different symbol for the switches in the figure.

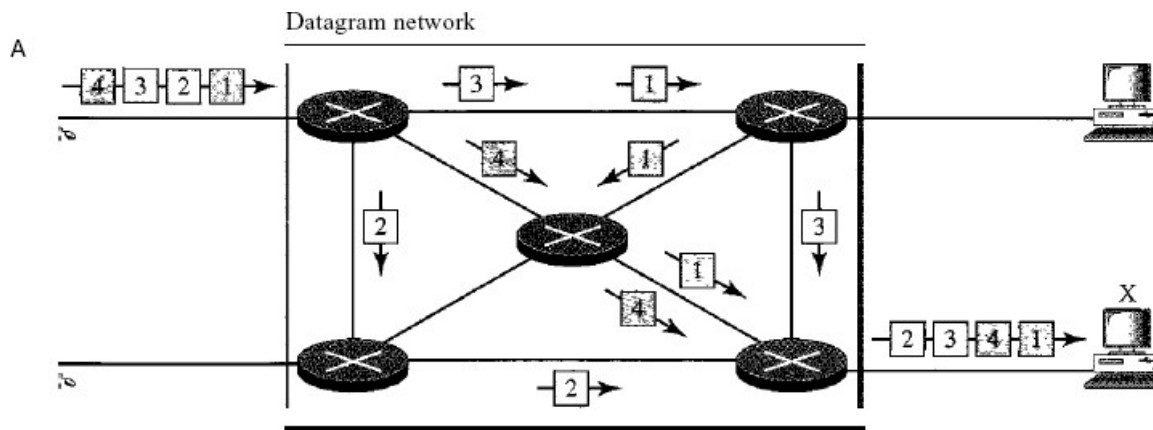


Fig: A datagram network with four switches (routers)

In this example, all four packets (or datagrams) belong to the same message, but may travel different paths to reach their destination. This is so because the links may be involved in carrying packets from other sources and do not have the necessary bandwidth available to carry all the packets from A to X. This approach can cause the datagrams of a transmission to arrive at their destination out of order with different delays between the packets. Packets may also be lost or dropped because of a lack of resources. In most protocols, it is the responsibility of an upper-layer protocol to reorder the datagrams or ask for lost datagrams before passing them on to the application.

The datagram networks are sometimes referred to as connectionless networks. The term *connectionless* here means that the switch (packet switch) does not keep information about the connection state. There are no setup or teardown phases. Each packet is treated the same by a

switch regardless of its source or destination.

Routing Table

If there are no setup or teardown phases, how are the packets routed to their destinations in a datagram network? In this type of network, each switch (or packet switch) has a routing table which is based on the destination address. The routing tables are dynamic and are updated periodically. The destination addresses and the corresponding forwarding output ports are recorded in the tables. This is different from the table of a circuit switched network in which each entry is created when the setup phase is completed and deleted when the teardown phase is over. Figure shows the routing table for a switch.

Destination address	Output port
1232	1
4150	2
9130	3

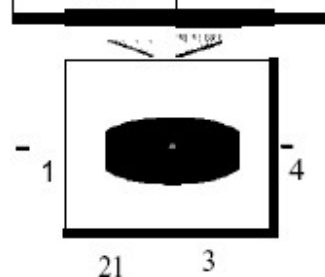


Fig: Routing table in a datagram network

Destination Address

Every packet in a datagram network carries a header that contains, among other information, the destination address of the packet. When the switch receives the packet, this destination address is examined; the routing table is consulted to find the corresponding port through which the packet should be forwarded. This address, unlike the address in a virtual-circuit-switched network, remains the same during the entire journey of the packet.

Efficiency

The efficiency of a datagram network is better than that of a circuit-switched network; resources are allocated only when there are packets to be transferred. If a source sends a packet and there is a delay of a few minutes before another packet can be sent, the resources

can be reallocated during these minutes for other packets from other sources.

Delay

There may be greater delay in a datagram network than in a virtual-circuit network. Although there are no setup and teardown phases, each packet may experience a wait at a switch before it is forwarded. In addition, since not all packets in a message necessarily travel through the same switches, the delay is not uniform for the packets of a message.

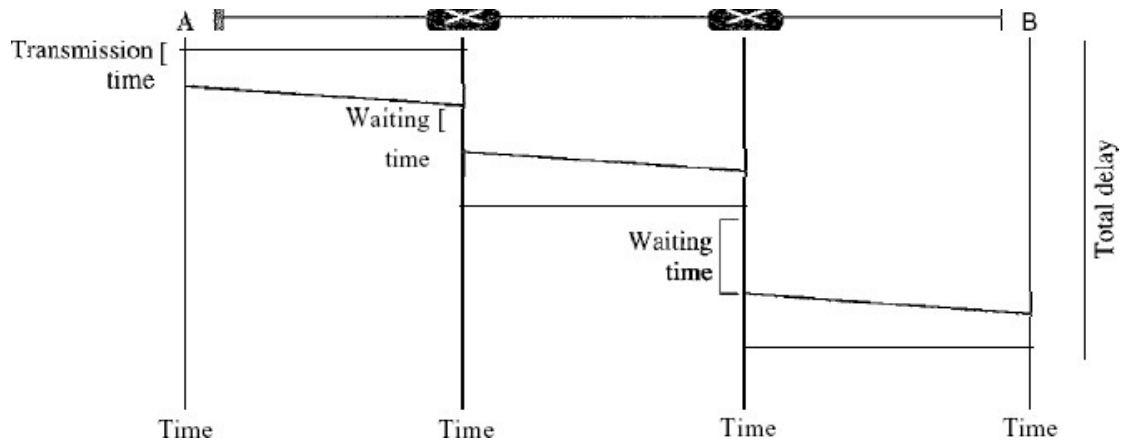


Fig: Delay in a datagram network

The packet travels through two switches. There are three transmission times ($3T$), three propagation delays (slopes $3t$ of the lines), and two waiting times ($W_1 + W_2$). We ignore the processing time in each switch. The total delay is

$$\text{Total delay} = 3T + 3t + W_1 + W_2$$

VIRTUAL-CIRCUIT NETWORKS:

A virtual-circuit network is a cross between a circuit-switched network and a datagram network. It has some characteristics of both.

1. As in a circuit-switched network, there are setup and teardown phases in addition to the data transfer phase.
2. Resources can be allocated during the setup phase, as in a circuit-switched network, or on demand, as in a datagram network.
3. As in a datagram network, data are packetized and each packet carries an address in the header. However, the address in the header has local jurisdiction (it defines what should be the next switch and the channel on which the packet is being carried), not end-to-end

jurisdiction. The reader may ask how the intermediate switches know where to send the packet if there is no

final destination address carried by a packet. The answer will be clear when we discuss virtual- circuit identifiers in the next section.

4. As in a circuit-switched network, all packets follow the same path established during the connection.

5. A virtual-circuit network is normally implemented in the data link layer, while a circuit-switched network is implemented in the physical layer and a datagram network in the network layer. But this may change in the future. Figure is an example of a virtual-circuit network. The network has switches that allow traffic from sources to destinations. A source or destination can be a computer, packet switch, bridge, or any other device that connects other networks.

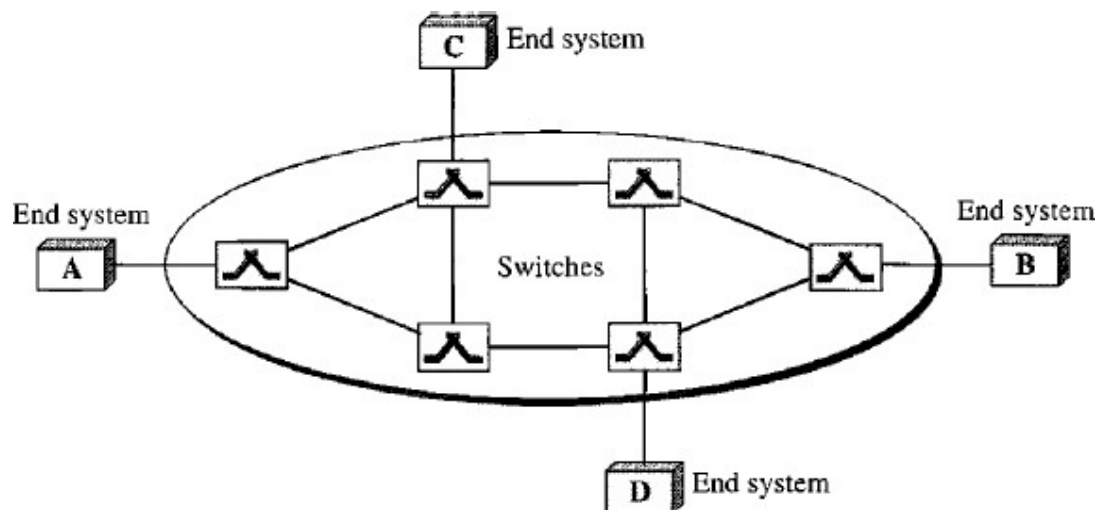


Fig a: *Virtual-circuit network*

Addressing

In a virtual-circuit network, two types of addressing are involved: global and local (virtual-circuit identifier).

Global Addressing: A source or a destination needs to have a global address—an address that can be unique in the scope of the network or internationally if the network is part of an international network. However, we will see that a global address in virtual-circuit networks is used only to create a virtual-circuit identifier, as discussed next.

Virtual-Circuit Identifier: The identifier that is actually used for data transfer is called the virtual-circuit identifier (Vci). A vci, unlike a global address, is a small number that has only switch scope; it is used by a frame between two switches. When a frame arrives at a

switch, it has a VCI; when it leaves, it has a different VCI. Figure 8.11 shows how the VCI in a data frame

changes from one switch to another. Note that a VCI does not need to be a large number since each switch can use its own unique set of VCIs.

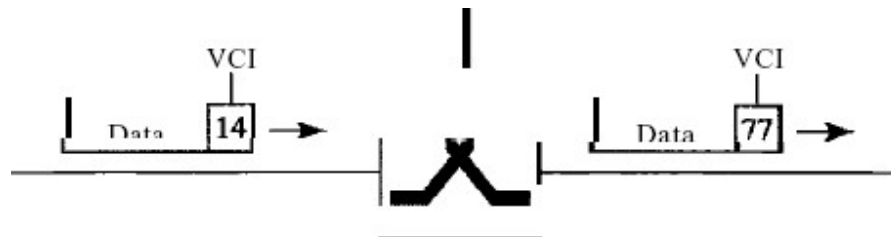


Figure 1 *Virtual-circuit identifier*

Three Phases

As in a circuit-switched network, a source and destination need to go through three phases in a virtual-circuit network: setup, data transfer, and teardown. In the setup phase, the source and destination use their global addresses to help switches make table entries for the connection. In the teardown phase, the source and destination inform the switches to delete the corresponding entry. Data transfer occurs between these two phases. We first discuss the data transfer phase, which is more straightforward; we then talk about the setup and teardown phases.

Data Transfer Phase

To transfer a frame from a source to its destination, all switches need to have a table entry for this virtual circuit. The table, in its simplest form, has four columns. This means that the switch holds four pieces of information for each virtual circuit that is already set up. We show later how the switches make their table entries, but for the moment we assume that each switch has a table with entries for all active virtual circuits. Figure 2 shows such a switch and its corresponding table. And also shows a frame arriving at port 1 with a VCI of 14. When the frame arrives, the switch looks in its table to find port 1 and a VCI of 14. When it is found, the switch knows to change the VCI to 22 and send out the frame from port 3. Figure 3 shows how a frame from source A reaches destination B and how its VCI changes during the trip.

Each switch changes the VCI and routes the frame. The data transfer phase is active until the source sends all its frames to the destination. The procedure at the switch is the same for each frame of a message. The process creates a virtual circuit, not a real circuit, between the source and destination.

Setup Phase

In the setup phase, a switch creates an entry for a virtual circuit. For example, suppose source A needs to create a virtual circuit to B. Two steps are required: the setup request and the acknowledgment.

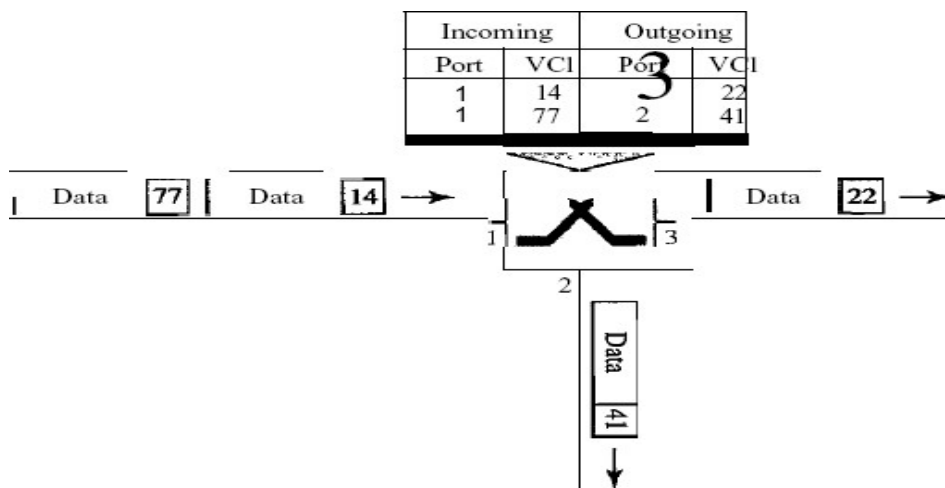


Figure 2 Switch and tables in a virtual-circuit network

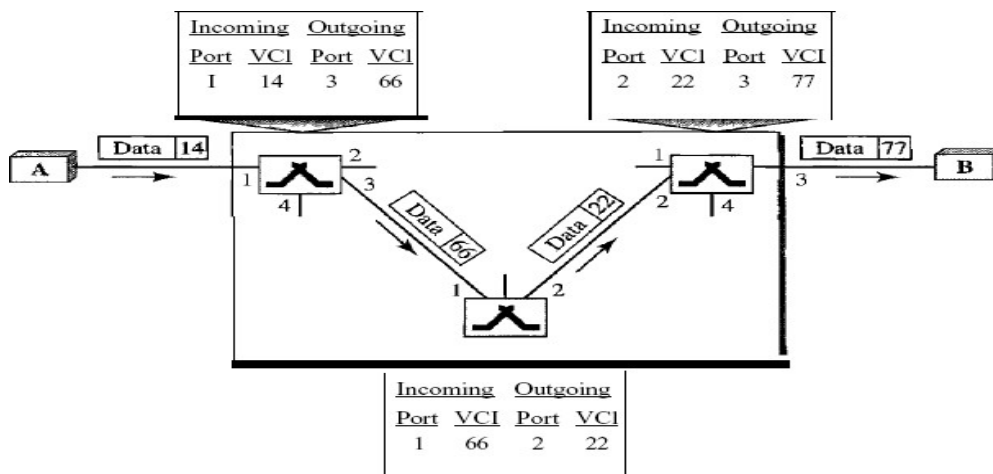


Figure 3 Source-to-destination data transfer in a virtual-circuit network

Setup Request

A setup request frame is sent from the source to the destination. Figure 4 shows the process.

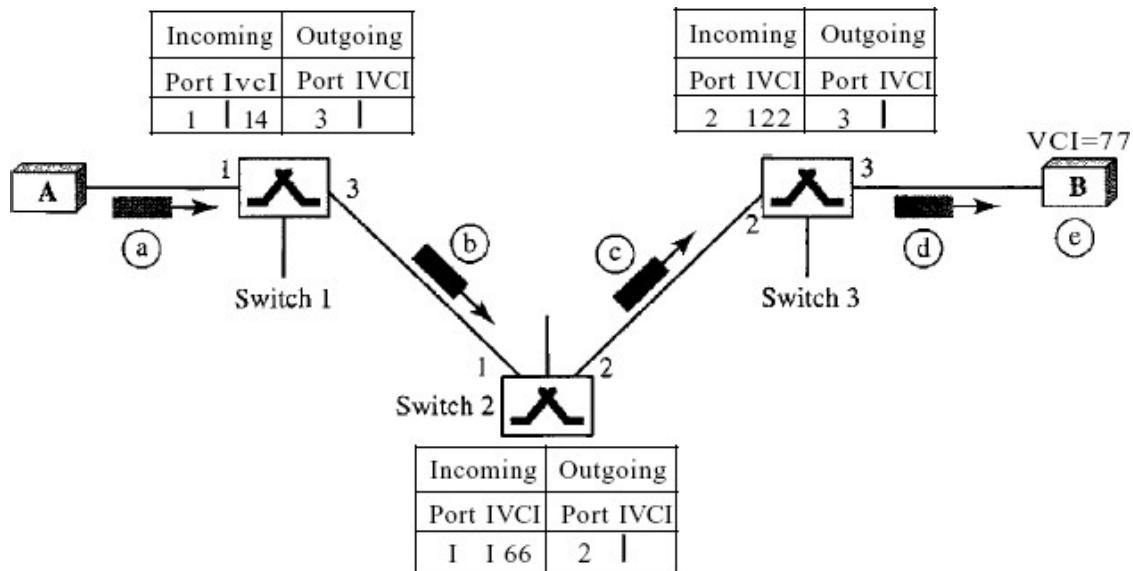


Figure 4 Setup request in a virtual-circuit network

- Source A sends a setup frame to switch 1.
- Switch 1 receives the setup request frame. It knows that a frame going from A to B goes out through port 3. How the switch has obtained this information is a point covered in future chapters. The switch, in the setup phase, acts as a packet switch; it has a routing table which is different from the switching table. For the moment, assume that it knows the output port. The switch creates an entry in its table for this virtual circuit, but it is only able to fill three of the four columns. The switch assigns the incoming port (1) and chooses an available incoming VCI (14) and the outgoing port (3). It does not yet know the outgoing VCI, which will be found during the acknowledgment step. The switch then forwards the frame through port 3 to switch 2.
- Switch 2 receives the setup request frame. The same events happen here as at switch 1; three columns of the table are completed: in this case, incoming port (1), incoming VCI (66), and outgoing port (2).
- Switch 3 receives the setup request frame. Again, three columns are completed: incoming

port (2), incoming VCI (22), and outgoing port (3).

Destination B receives the setup frame, and if it is ready to receive frames from A, it assigns a VCI to the incoming frames that come from A, in this case 77. This VCI lets the destination know that the frames come from A, and not other sources.

Acknowledgment A special frame, called the acknowledgment frame, completes the entries in the switching tables. Figure 8.15 shows the process.

a. The destination sends an acknowledgment to switch 3. The acknowledgment carries the global source and destination addresses so the switch knows which entry in the table is to be completed. The frame also carries VCI 77, chosen by the destination as the incoming VCI for frames from

A. Switch 3 uses this VCI to complete the outgoing VCI column for this entry. Note that 77 is the incoming VCI for destination B, but the outgoing VCI for switch 3.

b. Switch 3 sends an acknowledgment to switch 2 that contains its incoming VCI in the table, chosen in the previous step. Switch 2 uses this as the outgoing VCI in the table.

c. Switch 2 sends an acknowledgment to switch 1 that contains its incoming VCI in the table, chosen in the previous step. Switch 1 uses this as the outgoing VCI in the table.

d. Finally switch 1 sends an acknowledgment to source A that contains its incoming VCI in the table, chosen in the previous step.

e. The source uses this as the outgoing VCI for the data frames to be sent to destination B.

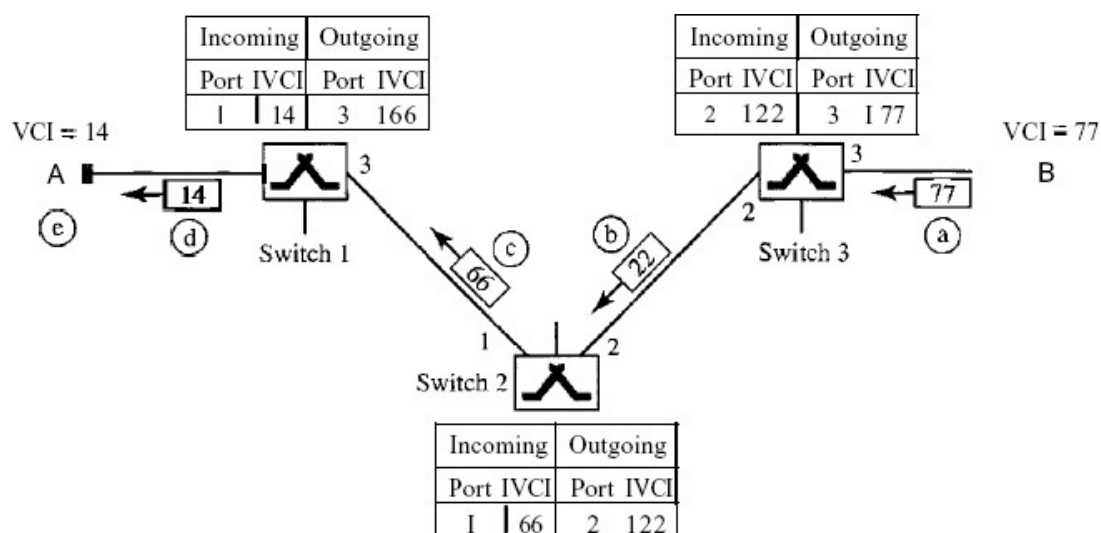


Figure 5 Setup acknowledgments in a virtual-circuit network

Teardown Phase

In this phase, source A, after sending all frames to B, sends a special frame called a *teardown request*. Destination B responds with a teardown confirmation frame. All switches delete the corresponding entry from their tables.

Efficiency

As we said before, resource reservation in a virtual-circuit network can be made during the setup or can be on demand during the data transfer phase. In the first case, the delay for each packet is the same; in the second case, each packet may encounter different delays. There is one big advantage in a virtual-circuit network even if resource allocation is on demand. The source can check the availability of the resources, without actually reserving it. Consider a family that wants to dine at a restaurant. Although the restaurant may not accept reservations (allocation of the tables is on demand), the family can call and find out the waiting time. This can save the family time and effort.

Delay in Virtual-Circuit Networks

In a virtual-circuit network, there is a one-time delay for setup and a one-time delay for teardown. If resources are allocated during the setup phase, there is no wait time for individual packets. Below Figure shows the delay for a packet traveling through two switches in a virtual- circuit network.

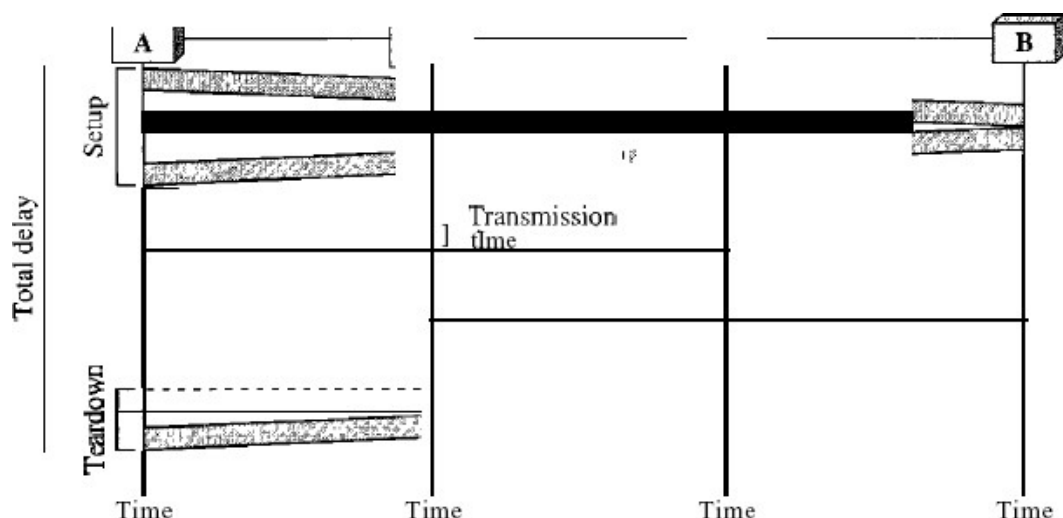


Fig: Delay in a virtual-circuit network

The packet is traveling through two switches (routers). There are three transmission times ($3T$), three propagation times ($3t$), data transfer depicted by the sloping lines, a setup delay (which includes transmission and propagation in two directions), and a teardown delay (which includes transmission and propagation in one direction). We ignore the processing time in each switch. The total delay time is

$$\text{Total delay} = 3T + 3t + \text{setup delay} + \text{teardown delay}$$